ABRIDGED CHAIN OF THOUGHT FOR IMPROVED SHORT GAPS BETWEEN PRIMES

The first target is to lower the bounded-gap constant from 246 to 244. It is tempting to regard this as a small numerical improvement, but the quantity that the sieve controls is not the length of an interval directly. It controls how many members of an admissible tuple can be prime together. I should first identify which integer threshold has to move.

Write

$$H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n),$$

and let $\mathrm{DHL}[k, 2]$ mean that every admissible k-tuple has infinitely many translates containing at least two primes. If such a tuple has diameter D, this implies $H_1 \leq D$: any two primes in the translate have consecutive primes between them, and one of those consecutive gaps is at most their distance. Admissibility means that the tuple misses a residue class modulo every prime. It is a finite check for a fixed tuple, since a set with k elements cannot occupy all the classes modulo a prime larger than k. Write $H(k)$ for the minimum diameter of an admissible k-tuple.

The existing 246 argument uses $k = 50$. Removing one coordinate suggests $k = 49$, and the tuple tables give something better than the requested 244: there is an admissible 49-tuple of diameter 240. Thus a genuine proof of $\mathrm{DHL}[49, 2]$ would settle the original target with room to spare. This is useful, but it does not make the analytic step any easier. The difference between 49 and 50 is exactly where a strict variational inequality stops being available.

For the ordinary Maynard simplex, put

$$\mathcal{R}_k = \{\mathbf{t} \geq 0 : t_1 + \cdots + t_k \leq 1\}, \qquad I(F) = \int F(\mathbf{t})^2\, d\mathbf{t},$$

and

$$J_i(F) = \int \left(\int F(\mathbf{t})\, dt_i\right)^2 d\mathbf{t}_{-i}.$$

The basic optimization is $M_k = \sup_F \sum_i J_i(F)/I(F)$. With a level of distribution approaching $1/2$, two primes require a quotient strictly larger than 4. A numerical search on this ordinary simplex cannot solve my problem, however ambitious the polynomial basis becomes. The published upper bound

$$M_k \leq \frac{k}{k-1}\log k$$

is already below 4 at both $k = 49$ and $k = 50$. I need to use the actual improvement in the Polymath argument, rather than optimize the wrong functional [1, 2].

The relevant device enlarges the support and truncates the prime marginals. Set

$$S = 1 + \varepsilon, \qquad T = 1 - \varepsilon,$$

allow F to have support in $\{\sum t_i \leq S\}$, and define

$$J_{i,T}(F) = \int_{\sum_{j\neq i} t_j \leq T} \left(\int_0^\infty F(\mathbf{t})\, dt_i\right)^2 d\mathbf{t}_{-i}.$$

The published calculation gives

$$M_{50,1/25} > 4.0043.$$

Its hypothesis is precise: under $\mathrm{EH}[\vartheta]$, the quotient must exceed $2/\vartheta$, with $(1+\varepsilon)\vartheta < 1$. Bombieri–Vinogradov provides every fixed $\vartheta < 1/2$, not the endpoint itself. Nevertheless a fixed rational trial with quotient $4+\eta$, for some $\eta > 0$, permits a sufficiently small fixed retreat from $1/2$. That is enough. There is no need to assume Elliott–Halberstam at its conjectural range [2].

This leaves a concrete question: can the enlarged-simplex quotient exceed 4 in dimension 49? The original paper explicitly leaves open the possibility that additional numerical work could lower the dimension. That is a much more promising starting point than an unspecified search for a new distribution theorem.

Before spending all the effort there, I should dispose of the attractive combinatorial shortcuts. Suppose the 50-tuple has endpoints 0 and 246, and write X_h for the indicator that $n+h$ is prime. Then

$$Y = \sum_h X_h - 1 - X_0 X_{246}$$

is positive whenever the translate contains two primes other than the sole endpoint pair. This is exactly the event needed to improve the extreme gap. But the existing sieve estimates only the first moments. The extra term is a weighted prime-pair correlation. Subtracting it in an identity does not estimate it.

There is a useful logical check here. I can construct a sparse set of integers that satisfies the qualitative conclusion of the tuple theorem while every gap in the set is at least 246. Enumerate the admissible systems and, very far apart, plant a pair of values for each system, using the Chinese remainder theorem to avoid the first several primes. The selected pair can always be separated by at least 246 when no admissible 50-tuple fits into a shorter interval. Repeating each system infinitely often gives all the required qualitative assertions. Even the prime number theorem and equidistribution in each fixed arithmetic progression can be retained: remove the primes belonging to a pair of distance at most 244, a set of size $O(x/\log^2 x)$ by the upper-bound sieve, and make the later plantings sufficiently sparse. This is not a model for the full analytic hypotheses. It shows precisely why the qualitative tuple conclusion alone cannot select a better pair.

The stronger different-bin statements have the same limitation. If every cross-bin pair of a large admissible tuple were within 244, its whole span would be at most 488. Avoiding one class modulo 2, 3, and 5 leaves at most eight positions in each block of thirty. An interval of 489 integers therefore contains at most 137 members of an admissible tuple. Large enough tuples necessarily have a distant cross-bin pair, which can again be planted. Thus the established partition theorems do not automatically force the particular short pair I want.

A weighted version is also expensive. For symmetric weights, a Hoeffding decomposition on the box $[0,S]^k$ compares the first and second coordinate averages. If

$$R = \frac{\sum_i J_{i,T}(F)}{I(F)},$$

then the normalized double marginal satisfies the lower bound

$$\frac{\int (\iint F\, dt_i\, dt_j)^2}{I(F)} \geq \frac{R(R-S)}{k(k-1)}$$

in the relevant range. At $k=50$, $S=1.04$, and $R=4.0043$, the right side is about 0.004845, already larger than the available 0.0043 surplus. This is unfavorable even before paying the actual upper-sieve constant and the additional modulus support. A better unweighted twin-prime constant would not, by itself, provide the corresponding weighted estimate. The endpoint route has acquired a real analytic obstruction, not just an inconvenient calculation.

I can make the enlarged-simplex calculation more explicit. A symmetric basis built from the radial variable $s = \sum t_i$ and power sums such as $p_2 = \sum t_i^2$ has exact simplex moments. For example, a basis element $(S-s)^a p_2^b$ has a marginal obtained by expanding $p_2 = p_{2,-i} + t_i^2$ and using

$$\int_0^{S-u} t_i^{2r} (S-u-t_i)^a\, dt_i = \frac{(2r)!a!}{(a+2r+1)!}(S-u)^{a+2r+1}.$$

This reduces both quadratic forms to factorials and polynomial coefficients. Arbitrary precision helps separate ill conditioning from a genuine small eigenvalue gap. Exact rational coefficients can ultimately replace a floating eigenvector.

The simplest angular family is inadequate. Powers of p_2 alone settle near 3.67 in dimension 49. Increasing its radial degree does not recover the missing three tenths. The full even-signature basis is much better: the exploratory sequence rises through roughly 3.91, 3.94, 3.96, and 3.98 as the degree increases. Odd signatures add a few thousandths, but extrapolation is not a certificate. Product profiles with a flexible radial factor also get close to 3.99 without clearing 4. At this point numerical improvements have to be judged against the remaining deficit, rather than against the immediately preceding run.

The finite-dimensional optimization also needs a useful scale for its errors. If A and B are the matrices of the prime and mass forms, the question is whether there is a vector v with

$$v^{\mathsf{T}}(A - cB)v > 0.$$

A large generalized eigenvalue computed from an ill-conditioned Gram matrix is not enough. The basis functions can be almost linearly dependent while their coefficients are enormous. It is safer to use the floating calculation to locate a direction, replace that direction by a rational vector, and evaluate its two quadratic forms directly. A tiny strict inequality survives only if the rounding error, basis error, and support error are all smaller than the margin. The successive degree calculations are useful diagnostics of the trial space, but there is no valid inference that their apparent limiting value is the true optimum.

I also considered an operator residual. If K is the positive operator whose Rayleigh quotient is the sieve ratio, then

$$\lambda_{\max}(K) \geq \frac{\langle F, K^2 F\rangle}{\langle F, KF\rangle}.$$

A trial can therefore reveal an improvement outside its own polynomial span through the size of KF. This is an efficient way to ask whether adding more of the same basis is worthwhile. It still requires an actual new trial or a rigorous moment bound at the end. An estimated residual variance cannot serve as the final strict inequality.

There is an obvious support improvement. A point contributes to the ith marginal only if

$$s - t_i \leq T.$$

Consequently every numerator is unchanged if I replace F by its restriction to

$$\Omega = \{s \leq S : \text{some } i \text{ has } s - t_i \leq T\}.$$

The deleted region is

$$D = \{T < s \leq S : \max_i t_i < s - T\}.$$

It contributes only to the denominator. Cutting it out is always legitimate. In fact, the original discussion already identifies this as the natural support, so I should not treat the observation itself as a new theorem. Its value is computational: it tells me where polynomial mass is being wasted.

For radial trials the deleted mass can be integrated by inclusion–exclusion. At a fixed total $s > T$, the density of the union where at least one coordinate is at least $s - T$ is proportional to

$$\sum_{m=1}^{k} (-1)^{m+1} \binom{k}{m} (mT - (m-1)s)_+^{k-1}.$$

Multiplying this by a radial polynomial gives exact one-dimensional integrals. The constant function loses a noticeable volume in the dead region, but that is misleading: its original quotient is poor. Good trials already suppress much of that region. The denominator saving has to be evaluated with the optimized weight.

Could a signed extension reach much farther out? The sieve argument suggests an enlarged support radius approaching 2 if the high marginals vanish exactly. The condition is

$$\int F(\mathbf{t})\,dt_i = 0 \quad \text{whenever} \quad \sum_{j\neq i} t_j > 1+\varepsilon.$$

A pure product times a radial function cannot exploit this. The condition becomes a Volterra equation, and a nonvanishing endpoint kernel forces the radial tail to vanish. Threshold-count features do allow coupled cancellation equations. They produce genuine signed extensions and some numerical gain, but not enough.

The exact constructions clarify the difficulty. If the number of coordinates above a threshold is r, the cancellation equations can be solved backwards between adjacent slabs. Their coefficients grow like binomial coefficients. The resulting tail has a large squared norm and very small correlation with the useful inner trial. A formally available extension need not be an efficient extension. I also have to distinguish a midpoint discretization from a continuum construction: cancellation at finitely many sampled sums does not establish cancellation on every fiber.

The other distribution results require the same discipline. Smooth moduli, well-factorable weights, a fixed residue class, and a general collection of Chinese remainder classes are different hypotheses. A small improvement in a displayed exponent is not useful if the modulus produced by the sieve falls outside the theorem. Prime minorants add a second issue. The inequality

$$A^2 \geq 2AB - B^2$$

can be multiplied by a nonnegative prime indicator. A minorant which is negative on composites does not preserve that argument when $2AB - B^2$ can also be negative. The attractive exponent must be accompanied by the right sign and the right support [3, 4].

One tempting smooth-modulus construction makes the issue particularly clear. Restricting each divisor coordinate to $d_i \leq x^\delta$ guarantees smoothness, but it also restricts every physical fiber to length δ. Cauchy–Schwarz then gives a total prime score at most $k\delta$. Moving to a larger distribution exponent usually forces δ down, so the gain in the exponent can be overwhelmed by the loss in available fibers. Replacing the hard coordinate cap by genuinely smooth divisors softens this loss, through the Dickman measure, but does not remove it.

For the soft version I can look for a dual upper bound rather than merely fail to find a good trial. A weighted Cauchy inequality reduces the many-variable ratio to one-dimensional integrals involving the Dickman density. Introducing a linear multiplier for the sum constraint makes the remaining expression affine in an auxiliary variable, so its maximum occurs at an endpoint. This reduces a proposed uniform obstruction to elementary logarithmic bounds in one variable. The distinction matters: a poor product trial is evidence about that trial, while a verified dual majorant would exclude every function in the specified weighted space.

The hybrid unrestricted/smooth construction likewise has an exact Gram matrix. If G represents the unrestricted part, H the smooth part, and $w = \prod_i w_\xi(t_i)$, where $w_\xi(t) = \rho_D(t/\xi)$ and ρ_D is the Dickman function, the mass is

$$\int (G^2 + 2wGH + wH^2).$$

It is not $\int G^2 + \int wH^2$. For prime marginals, G is integrated against ordinary measure and H against the Dickman measure. The useful regions depend on whether a mixed pair is safely below level $1/2$ or whether both members are smooth. This explains why adding a small smooth shell to a successful unrestricted trial is not automatically a positive improvement. The cross correlation and the permitted modulus range have to be optimized together.

The important change is to stop equating densely divisible with smooth. A number is y-densely divisible if it has a divisor within a factor y of every scale up to its size. Smoothness implies this, but the converse is very far from true. The Polymath distribution statement permits prime factors in an arbitrary bounded prime set; its dense-divisibility condition does not require every prime factor to be at most y. That distinction removes the coordinate cap which had been destroying the variational quotient [5].

For a squarefree integer with increasing prime factors $p_1 < \cdots < p_r$, single dense divisibility is equivalent to

$$p_j \le y \prod_{\ell<j} p_\ell \qquad (1 \le j \le r).$$

Necessity follows by looking at the gap before p_j. Sufficiency follows by adjoining the primes in order. More generally, adjoining a block $m \le yN$ to a y-dense integer N preserves single dense divisibility. This means that I can certify density from the coordinate divisor blocks, without factoring them individually.

Let their normalized logarithms, in increasing order, be $t_{(1)}, \dots, t_{(k)}$, and define

$$C(\mathbf{t}) = \max_j \left(t_{(j)} - \sum_{\ell<j} t_{(\ell)} \right), \qquad s = \sum_j t_j.$$

If $C(\mathbf{t}) \le \xi$, the product of the blocks is R^ξ-dense. The crucial statistic is not C alone, but

$$H(\mathbf{t}) = s + C(\mathbf{t}) = \max_r \left(\sum_{j<r} t_{[j]} + 2t_{[r]} \right),$$

where the brackets indicate decreasing order. Each expression on the right increases when a coordinate increases. Hence H is coordinatewise monotone, even though the original increasing-order definition of C does not make that property obvious.

I can now use a buffered domain

$$E_c^{(k)} = \{\mathbf{t} \ge 0 : s \le c \text{ or } H(\mathbf{t}) \le c + \xi\}.$$

It is downward closed. Above its unrestricted core $s \le c$, it gives

$$C(\mathbf{t}) \le c + \xi - s \le \xi,$$

with a stronger density parameter as the total size increases. Downward closure matters because the sieve coefficient is a primitive of F:

$$f(\mathbf{t}) = \int_{\mathbf{u} \ge \mathbf{t}} F(\mathbf{u})\, d\mathbf{u}.$$

A good support condition for F would be useless if taking this primitive introduced forbidden divisor tuples. Here it does not.

Write $R = x^\rho$ and $B = 1/(2\rho)$. For outer radius S and inner radius T, choose

$$a = B - T, \qquad b = B - S,$$

and restrict the outer and inner variables to

$$\mathcal{D} = E_a^{(k)} \cap \{s \le S\}, \qquad \mathcal{L} = E_b^{(k-1)} \cap \{s \le T\}.$$

Consider a mixed modulus formed from an outer divisor root and an inner one. If their total logarithmic size is at most B, ordinary Bombieri–Vinogradov applies. Otherwise both roots have left their unrestricted cores, so both are densely divisible. The single-density least-common-multiple closure then supplies the stronger distribution estimate. This is a support theorem matched to the actual mixed modulus, not an assumed improvement for arbitrary moduli.

The source inequalities can be checked with a small exact choice. Taking

$$\omega = \frac{1}{200}, \qquad \delta = \frac{61}{4000}, \qquad \sigma = \frac{1001}{10000}$$

gives a singly dense distribution exponent $1/2+2\omega = 51/100$. Among the necessary inequalities are

$$54\omega + 15\delta + 5\sigma < 1, \qquad 68\omega + 14\delta < 1, \qquad \sigma > \frac{1}{18} + \frac{28}{9}\omega + \frac{2}{9}\delta.$$

For these parameters the first two left sides are 3997/4000 and 1107/2000. There is genuine strict slack. The residue issue also has to be carried through: the Chinese remainder choices are averaged in the same coherent global-residue form used by the published sieve argument. A theorem about one arbitrary but coherent residue class is not permission to choose an unrelated class for every modulus.

The variational target is now

$$\rho \, \frac{\sum_i \int_{\mathcal{L}} \left(\int F \, dt_i \right)^2}{\int_{\mathcal{D}} F^2} > 1.$$

The necessary raw quotient is below 4. The density masks cost something, but numerical experiments suggest that they cost less than the gain in ρ. This is the first route with a plausible quantitative margin.

A convenient trial is positive and separable apart from a radial polynomial. On a rational grid, use

$$g_j = \frac{1}{1+195(j+\frac{1}{2})h}, \qquad f_r = 1 - \frac{3}{2}d_r - 12d_r^2 - 13d_r^3, \qquad d_r = (r+k/2)h - \frac{9}{10}.$$

The function is constant on each coordinate box and equals $f_{j_1+\cdots+j_k}\prod_i g_{j_i}$. Its positivity is checked over the entire grid, not inferred from selected samples. For the first 49-dimensional certificate,

$$S = \frac{26}{25}, \quad T = \frac{24}{25}, \quad h = \frac{13}{150000}, \quad \xi = \frac{61}{1020}.$$

All admissibility tests use the upper corner of a box. The midpoint is only where the box's constant amplitude is defined. This distinction avoids falsely counting a box which crosses a forbidden boundary.

The full box sums reduce to polynomial convolutions. The remaining masked losses can be bounded by assigning a canonical offending coordinate and applying positive Chernoff majorants. These need not involve transcendental constants: for an integer event $U \geq K$ and a rational $z > 1$,

$$\mathbf{1}_{\{U\geq K\}} \leq z^{U-K}.$$

The resulting finite calculation uses rational data and outward-rounded arithmetic. It is particularly helpful to separate an exact definition of every quantity from the means of evaluating it. A positive numerical eigenvalue is evidence for a candidate; an enclosure with a positive lower endpoint is the finite inequality required by the argument.

The full sums have a transparent exact form. Put

$$P(z) = \sum_{j\geq 0} g_j^2 z^j.$$

For the unmasked radial cutoff $\sum j_i \leq M$, the mass is

$$I_0 = h^k \sum_{r\leq M} f_r^2 [z^r] P(z)^k.$$

For a fixed prime coordinate, the other coordinates have total index r, and its marginal is $h\sum_{j\leq M-r} g_j f_{r+j}$ times their product weight. Hence, with the inner full-box cutoff Q,

$$J_0 = kh^{k+1} \sum_{r\leq Q} [z^r] P(z)^{k-1} \left(\sum_{j\leq M-r} g_j f_{r+j} \right)^2.$$

Angular polynomials add marked moments to these convolutions, but the principle is unchanged. The forty-nine-dimensional integral has become a finite coefficient identity. This is much easier to audit than a black-box quadrature in forty-nine variables.

Masking has opposite effects on the two sides of the quotient, so the inequality directions must be explicit. To upper-bound the masked denominator, I may subtract a rigorously lower-bounded subset of the removed mass from I_0. To lower-bound the numerator, I must subtract an upper bound for every lost contribution. For a nonnegative trial, if a fiber marginal changes from A to $A-\Delta$, then

$$(A-\Delta)^2 = A^2 - 2A\Delta + \Delta^2 \geq A^2 - 2A\Delta.$$

This permits positive upper bounds for the loss. It does not permit me to subtract a lower estimate of that loss, or to discard a residual because it looks small in random samples.

A canonical offender makes union bounds less wasteful. Instead of charging a bad box once for every coordinate that violates the dense condition, charge it to a specified largest or first violating coordinate. Large-coordinate and two-coordinate events can then be summed relatively sharply. The unresolved configurations receive a separate positive majorant. A small exact-grid example is valuable here: it can enumerate every box and compare the direct masked sum with the convolution identity and the majorant. Agreement of those identities checks the algebra; the high-dimensional enclosure then checks their actual quantitative use.

The calculation gives the enclosed quotient

$$3.926140714916510577 3\ldots.$$

Using the strict exponent $\rho_* = 25499/100000$, slightly below $51/200$, its physical score is

$$1.0011266208965610321\ldots > 1.$$

The retreat simultaneously makes the Bombieri–Vinogradov boundary and the enhanced-distribution boundary strict. I now have a proposed argument for DHL$[49,2]$, and the independently checked 49-tuple converts that into diameter 240. The useful lesson is not merely the decimal surplus. A buffered, hereditary support has made a restricted distribution theorem useful to a multidimensional sieve.

The next stage starts from a stronger intermediate construction in dimension 43, with diameter 200, and asks whether dimension 42 can be made to work. A 43-tuple cannot be squeezed into diameter 198, so there is again no substitute for lowering the dimension. The initial 42-dimensional masked score is about 0.9928. I need another structural saving rather than a tiny adjustment of the existing parameters.

The first saving is asymmetric. If the outer root D is y-dense and every prime factor of the inner root E is at most yD, then adjoining those primes preserves density of $\mathrm{lcm}(D,E)$. Therefore both roots need not be dense. In the relevant outer shell, the lower bound on D is already larger than the upper bound on E, making the prime-size condition automatic. For single-density distribution, I can put the expensive density restriction on the outer root alone.

Before using the double-density theorem, I need the recursive definition. Order zero imposes no condition. An integer n is strongly r-fold Y-densely divisible if, for every split $j+k=r-1$ and every target scale $X \in [1, Yn]$, there is a factorization $n = ab$ with $X/Y \leq b \leq X$, where a has order j and b has order k at the same parameter Y. At order one this is ordinary dense divisibility. The higher orders require all these recursive factorizations, not just a dense list of divisors [5].

A more substantial gain comes from the double-density theorem. It is not legitimate to say that the least common multiple of two singly dense integers is automatically doubly dense. The shared factors and the available scales have to be accounted for. The useful statement is more precise. If D, E are squarefree and singly y-dense, and

$$P^+(D) \leq yE, \qquad P^+(E) \leq yD,$$

then, with $G = (D, E)$,

$$\mathrm{lcm}(D, E) \quad \text{is strongly doubly } yG\text{-densely divisible.}$$

The factor G is a real loss, not notation which can be ignored.

To see the mechanism, merge the distinct prime factors of D and E in increasing order. Before the next prime p, let D_p, E_p be the two prefixes and $G_p = (D_p, E_p)$. The accumulated modulus is $N = D_p E_p / G_p$. The prefixes inherit single dense divisibility. A balanced insertion lemma says that if $a \mid N$ is singly Y-dense and

$$p \le Y \min(a, N/a),$$

then adjoining p preserves strong double Y-density. The next prime in the other root supplies the comparison with the complementary factor; if that root is exhausted, the displayed prime-size condition supplies it. This proves the transfer with $Y = yG$, including overlaps.

The buffered domains were already built to pay for exactly this gcd. If the actual least common multiple lies above a previous distribution level R^B, then

$$G < \frac{DE}{R^B}.$$

The outer buffer gives density parameter $R^{a+\xi}/D$, and $a = B - T$. Consequently

$$\frac{R^{a+\xi}}{D}\, G < R^{\xi} \frac{E}{R^T} \le R^{\xi}.$$

The inner calculation is identical. The apparent gcd loss disappears against the radial slack. This is stronger than imposing a fixed dense parameter on both roots and then hoping their overlap is small.

The double-density source permits

$$56\omega + 16\delta + 4\sigma < 1$$

along with the common Type II and Type III conditions. Near $\sigma = 1/10$, its limiting ω is $3/280$, larger than the useful single-density range. A sequence of source levels can cover the mixed moduli: below one level use the preceding theorem; above it activate the next buffered restriction. Single-density levels restrict only the outer root, while double-density levels use the paired transfer.

The source ladder is more than an interpolation of exponents. Each level has a different permitted density parameter. A mixed product can be large while its least common multiple is smaller because of a common factor. I have to classify it by the actual modulus whenever invoking a distribution statement, and use the product size only to bound what that common factor could be. If a modulus falls into an earlier band, that earlier theorem has a larger density allowance; this can absorb the extra overlap. If it falls in the last band, the small width of the band supplies the overlap bound. Treating all moduli as though they were coprime would miss exactly the configurations which the paired argument was designed to handle.

This also explains why a nearly equal sequence of source thresholds can be useful. The first theorem pays for a substantial unrestricted core. Later theorems only control increasingly thin shells. A terminal parameter close to the edge of the published range can have a very small density allowance without forcing that small allowance on the entire trial. The correct object to optimize is the whole sequence of cores and shells. Its costs depend on where the trial puts its mass, not just on the final distribution exponent.

At this stage exact discretization becomes part of the mathematical argument. Several nominally different source thresholds can define identical integer masks. Comparing rounded real numbers may create a spurious shell. Conversely, replacing an exact floor inverse by an informal reciprocal can omit legitimate offenders. The cure is to express each violation as an integer inequality at the upper box corner, determine the first possible offender exactly, and only then merge equal masks. Ties between coordinates must belong to the relevant upper tail.

The positive trial now includes angular information, with its global positivity established by rational Bernstein subdivision. For the final 42-dimensional construction in this stage,

$$S = \frac{1041}{1000}, \qquad T = \frac{2397}{2500}, \qquad \rho_* = \frac{2606499}{10000000}.$$

Seven single-density levels and eight double-density levels reduce to two effective masked shells on the chosen grid. The numerator losses are separated into large-coordinate events, pair events, and the remaining residual events. This is useful because a loose bound for a rare residual can consume a margin which is only a few parts in ten thousand.

The completed enclosure is

$$\rho_* \frac{J_{\text{lower}}}{I_{\text{upper}}} \in [1.0005661567813 \pm 2.98 \cdot 10^{-14}].$$

Every finite component is included before drawing the conclusion. The result is a proposed DHL$[42, 2]$ argument, and the explicit admissible 42-tuple has diameter 196. Independent checks of the convolutions, source inequalities, integer event covers, and final interval sum are valuable here because they test different possible failure modes. Increasing numerical precision alone would not detect an omitted region. The interval arithmetic supplies the rounding control, while the finite combinatorial checks supply the coverage control [6].

For the next target, diameter 188, the tuple calculation is decisive: $H(41) = 188$, and no admissible 42-tuple fits. Thus the new analytic problem is DHL$[41, 2]$. The successful dimension-42 certificate does not automatically contain it.

A linear endpoint weighting makes the obstruction especially clear. Join two tuple positions by a bad edge when their separation exceeds 188. On the relevant diameter-196 tuple this graph is bipartite. A linear score which cannot be positive on a single prime or on a bad pair must obey

$$c_v \leq 1, \qquad c_u + c_v \leq 1 \quad \text{on every bad edge.}$$

The absence of a 41-point admissible window forces a matching of size at least two, and hence $\sum_v c_v \leq 40$. That bound is attained by dropping two suitable endpoints. Multiplying the existing symmetric score by $40/42$ puts it well below 1. Allowing negative coefficients does not repair the linear program. A nonlinear pair penalty returns to the weighted correlation problem encountered at the start.

It is also dangerous to extrapolate the density transfer by its name. Tests using the literal recursive definition produce counterexamples to naive claims that single plus double yields triple, or double plus double yields quadruple, even under tempting balance assumptions. The published higher-density conditions ask for particular recursive factorizations, not merely a large number of divisors. Any stronger transfer needs a new insertion argument with the exact power and gcd budget visible.

There are several honest support improvements worth trying. The prime-detected coefficient only sees the face $t_i = 0$. Its support is controlled by the projection of the full support onto the other coordinates. Thus a full trial can occupy

$$\sum_{j \neq i} t_j \leq U \quad \text{for every } i,$$

which is equivalent to $s - \min_i t_i \leq U$ and permits $s \leq kU/(k-1)$. The ordinary total-sum cutoff was stronger than the prime-face argument required. Nonprime mass still has ample support budget at these radii.

Signed tails give a further extension when an unsafe face marginal is made identically zero. On a region with a unique bad face, I can subtract the average along that fiber. If Ψ is the useful first-variation function, the best zero-mean direction on a fiber of length L is

$$H(u,t) = \Psi(u,t) - \frac{1}{L}\int_0^L \Psi(u,v)\,dv.$$

It has positive first variation unless Ψ was constant there. Disjoint exact examples show that this is a genuine variational improvement. In the high-dimensional trial, however, the available correlation is tiny: the first tested shell improves the score by only about $3 \cdot 10^{-8}$. The theorem is useful information about the support, but the measured scale rules it out as the immediate answer.

There is a further boundary check in the lifted construction. It is not enough to verify a vanishing marginal for the coordinate whose face is visibly bad. A tail which fixes that face can create another bad face. The support can be partitioned by the complete set of bad faces, and a derivative in every bad coordinate makes the corresponding marginals vanish. On a rectangular fiber, for example, a compactly supported potential with zero boundary values has a derivative whose integral is exactly zero. Products of such derivatives handle multiple bad coordinates. This gives a continuum construction rather than an identity which holds only at a set of discretization points.

The numerical limitation is then understandable. An extension far outside the original support has many bad faces, so its cancellation must occur in many directions. High-order oscillation makes its correlation with a smooth positive base very small. Even when an exact signed tail exists, its denominator can be large compared with its first-order gain. That is why a qualitative strict improvement does not settle the quantitative 41-coordinate deficit.

The density masks themselves can be weakened for small primes. A prime $p \le Y$ can be inserted harmlessly at every fixed density order. Only larger primes need the restrictive chain inequalities. This suggests an activated obstruction which ignores coordinates already below the source threshold. The double-density proof survives this change if I start with all primes at most Y, then insert the rough primes in order. For a rough prime in one root, the next rough prime in the other root supplies the balanced comparison; exhausted roots use the same cross cap as before. The gcd accounting remains essential. This is a valid enlargement for both roots, not merely a numerical relaxation.

Another option reverses the single-density orientation: make the inner root dense and put an adaptive maximum-prime cap on the outer root. It saves about 0.00032 for one tested profile. I then wondered whether correlating several orientations could collect their gains. For nested source domains, an exact triangular completion shows that the optimized score is a positive weighted average of the scalar scores. There is no hidden gain there. Crossed source ladders can make the rectangles incomparable, and a small exact model shows that correlated channels can then help in principle. The actual sieve domains overlap almost completely, however, and the tested crossed constructions remain below the best scalar orientation. The abstract counterexample to a no-gain statement is not itself a useful prime-gap construction.

The numerical work agrees on the scale of the remaining problem. Flexible positive angular trials, richer radial families, and a much finer valid source ladder bring the dimension-41 score to roughly 0.9964. Some trials look excellent with the masks omitted and fail badly once the source restrictions are restored. A gain found while choosing the coefficients must also survive a fresh evaluation. The missing few thousandths are too large to attribute to rounding and too small to ignore the arithmetic structure of the masks.

I returned to arithmetic restrictions on the divisor roots themselves. If all their prime factors are small, the logarithmic measure is no longer Lebesgue measure but a Dickman-weighted measure. Writing $w_\xi(t) = \rho_D(t/\xi)$, the natural forms become

$$I(F) = \int F^2 \prod_j w_\xi(t_j)\,d\mathbf{t},$$

$$J_i(F) = \int \left(\int F w_\xi(t_i)\, dt_i \right)^2 \prod_{j \neq i} w_\xi(t_j)\, d\mathbf{t}_{-i}.$$

The change of variables $z = \int_0^t w_\xi(u)\, du$ turns the measure into Lebesgue measure but changes the support. It does not restore the unrestricted simplex. Simple coordinate Cauchy bounds and more refined dual bounds explain why the available smooth-modulus exponents do not provide the needed score. Mixed unrestricted and smooth roots lead to nontrivial cross terms; they cannot be treated as independent positive contributions.

There was an instructive false start in the finite quadratic forms. For the ordinary divisor kernel $1/[d,e]$, adding a densely divisible divisor with a nontrivial small-prime part can be orthogonal to the optimized smooth residual. Rough-prime sector decompositions suggest an exact no-gain statement. But the prime face uses $1/\varphi([d,e])$, not $1/[d,e]$. Small exact examples show a genuine gain from adding rough factors. The finite equality was false and had to be withdrawn.

An asymptotic comparison may still hold after presieving: only boundedly many sufficiently large rough primes occur in a coefficient of bounded radial support, and the kernel discrepancy can be estimated on those sectors. The bounded-degree estimate is what distinguishes that statement from the false exact one. It also only addresses the seeded coordinate filters under consideration. It does not rule out a restriction on the full product of the coordinate divisors. That distinction now looks important.

The coordinate-block certificate throws away information. A large block may consist of several prime factors, and its primes can interleave with the primes in the other blocks. Ordering the block sizes treats all those factors as though they had to be inserted at once. Instead, consider the actual prime factors of the complete root D. A buffered condition

$$D \text{ is } (K/D)\text{-densely divisible}$$

has an unexpectedly clean form. With decreasing prime factors $p_1 > \cdots > p_r$, it is equivalent to

$$p_1 \cdots p_{j-1} p_j^2 \leq K \qquad \text{for every } j.$$

Deleting prime factors only makes these inequalities easier. Thus this global buffered support is hereditary, although density at one fixed parameter is not hereditary. It resembles the support inequalities of a beta sieve and preserves exactly the feature needed by the Selberg primitive.

Finite symmetric calculations show why this is worth pursuing. In a 41-coordinate example with the correct hereditary buffers, replacing the coordinate-block condition by the exact global-prime condition raises the true prime-face quotient by a factor of approximately 1.00476. That is finally comparable with the missing few thousandths. It is not yet the asymptotic sieve calculation, but it identifies a specific loss in the old geometry.

The finite prime-factor experiment needs its own caution. Small prime sets can exaggerate effects which disappear after presieving. The correct comparison keeps the dimension fixed, excludes the small primes which belong in the pre-sieving modulus, respects the radial product cutoff, and uses the true totient kernel for the prime face. Symmetry reduces the number of coefficient states by grouping assignments with the same orbit, but that reduction must preserve the exact cross-coordinate coprimality conditions. A gain in a surrogate matrix without those conditions would not identify a usable sieve improvement.

Here the global buffered gain survives those corrections. The old coordinate support is genuinely contained in the new hereditary prime-factor support, and there are explicit additional states. Their benefit has a plausible interpretation: two different coordinate blocks can each look too large when regarded as indivisible objects, while their interleaved prime factors provide enough intermediate scales for the product. This is information which no polynomial in the coordinate totals alone can recover.

The continuum description should retain the prime factors rather than only their sum. Conditional on a logarithmic coordinate total t, the largest prime fragment has distribution governed by the Dickman function:

$$\Pr(\text{largest fragment} \le z) = \rho_D(t/z).$$

Equivalently, size-biased prime fragments are described by uniform stick breaking. Splitting the primes into bands gives a small-prime Dickman component and a collection of larger jumps with intensity du/u. The normalization identities are important: the total logarithmic coordinate must still have the correct harmonic measure. Sampling arbitrary fragments with the right-looking mean would change the sieve functional.

The first continuum experiments are mixed. One-sided global restrictions outperform putting the same restriction on both roots, but the measured score remains below 1, even when the unmasked trial is above it. This makes the next question sharper: can a stronger pair transfer use weaker global power-chain conditions on each root?

Here ordinary recursive density and a sufficient power-chain condition must be kept separate. Suppose the rough primes in a root satisfy

$$p^2 \le y_D\, D_{<p},$$

and the corresponding condition holds in the other root. If its next prime is $r \ge p$, then

$$p^4 \le y_D y_E\, D_{<p} E_{<p} = y_D y_E G_{<p} N_{<p}.$$

Thus an insertion criterion for fourth-order density becomes plausible when $y_D y_E G \le Y$. The adaptive budgets can be split between the two roots:

$$y_D = R^{a+\xi_D - s_D}, \qquad y_E = R^{b+\xi_E - s_E}, \qquad \xi_D + \xi_E = \xi.$$

Above the previous modulus level, the same gcd inequality used in the successful double-density argument pays for their product. Exhausted roots require additional maximum-prime caps. Those caps are cheap in the trial geometry, but they must still be included in the statement.

This does not revive the disproved slogan that two doubly dense numbers are automatically fourfold dense. It proposes a different sufficient hypothesis, expressed through rough-prime power chains and explicit budgets. The immediate task is to turn that hypothesis into a complete insertion proof and then into a finite calculation for the global prime-fragment support. A Monte Carlo surplus, should one appear, will still leave that last certification problem to solve.

The first forty-one-dimensional crossing is small enough that I have to understand exactly what is producing it. The forty-one-coordinate trial gives a quotient a little above one after the source restrictions are imposed. An independent evaluation of a fixed trial gives about 1.00052, and subsequent evaluations give comparable values. This is evidence that the support is worth pursuing. It does not yet tell me how to prove the inequality. The margin is only a few parts in ten thousand, and a plausible error estimate can easily be larger than the entire improvement.

I will use u for a prime's normalized logarithm and write

$$L_D(u) = \sum_{v \in D,\ v > u} v, \qquad H_{m,\xi}(D) = \max_{u \in D,\ u > \xi} (L_D(u) + mu).$$

Here D is the prime-fragment configuration of one divisor root, with total mass s_D; the maximum is zero if no fragment exceeds ξ. I will call an inequality imposed on the root containing a prime its owner condition. The order-two condition considered above is therefore an H_3 bound. If C is the relevant owner ceiling, a failure at a fragment u means

$$L_D(u) + 3u > C.$$

Equivalently, if $L_{<u}$ is the mass of smaller fragments,

$$L_{<u} < s_D + 2u - C. \tag{1}$$

The two descriptions suggest different ways to estimate the exceptional set. The first involves the large fragments, which are few. The second involves a lower tail of many small fragments, which ought to be rare.

The coordinate totals t_i alone do not specify this functional. Two integers can have the same normalized logarithm and very different prime partitions. In the denominator, the source indicator is evaluated on all forty-one fragment configurations. In a squared face marginal, the other forty configurations are shared, but the two copies of the integrated coordinate have independent conditional fragment partitions. Reusing one random fiber in both factors would compute the expectation of a square, instead of the square of the conditional expectation. This distinction has no visible effect in the ordinary unmasked Maynard integral, but it becomes essential once the support depends on fragments [1, 2].

The fixed trial family has a positive rational coordinate profile multiplied by a symmetric angular polynomial:

$$F(t) = \prod_{i=1}^{41} g(t_i) \sum_{\sigma,d} c_{\sigma,d}(s - 9/10)^d P_\sigma(t), \qquad s = \sum_i t_i.$$

Here F denotes the amplitude as a function of the coordinate totals; the actual trial also contains the indicator of the permitted fragment support. The eight angular signatures are $1, P_2, P_3, P_4, P_5, P_6, P_2^2, P_2P_3$, where $P_j = \sum_i t_i^j$, and the radial degrees run from zero through six. This gives fifty-six coefficients. It is a small enough family to freeze exactly, but rich enough that expanding its square produces substantial signed cancellation. The goal is to preserve that useful cancellation while bounding the measure on which the polynomial is evaluated.

The sign of the optimized polynomial initially looks inconvenient. After reversing its overall sign, almost all sampled values are positive, but a thin region near the outer radial boundary still contains negative values. Adding a known positive trial makes the sampled polynomial nonnegative, at some cost to the quotient. Constraining positivity at many sampled points costs surprisingly little, but then more deliberately chosen points reveal new negative values. This is a useful warning: positivity on the part of the simplex carrying most of the mass is not global positivity. A square of a lower-degree polynomial would settle the issue algebraically, but the square family tested here gives a quotient below one.

There is a simple variational observation. Replacing F by $|F|$ leaves

$$I(F) = \int F^2$$

unchanged and can only increase every squared one-coordinate marginal. A positive trial therefore exists whenever a signed trial works. The difficulty is evaluating the absolute value sharply. It is better to keep the signed polynomial and arrange the error estimate as a sum of positive squares.

For one face, let V be the marginal before a source restriction, and let B be the marginal removed by that restriction. Then

$$(V - B)^2 \geq V^2 - 2VB.$$

For every positive c, Young's inequality gives

$$2VF \leq cF^2 + c^{-1}V^2. \tag{2}$$

Consequently the loss can be bounded by positive integrals involving F^2 and V^2, even when F changes sign. The parameter c can depend on the part of the exceptional set being bounded. There is no reason to insist that the same value be optimal for every fragment size. This eventually matters more than trying to prove positivity of the original polynomial.

I also need to use all the room in the arithmetic source. Put $a = B_0 - T$ and $b = B_0 - S$ for the two safe cores of a modulus band with lower endpoint B_0. If $Q = [D, E] > R^{B_0}$ and $G = (D, E)$, then the adaptive density scales satisfy

$$y_D y_E G = \frac{R^{a+b+\eta_D+\eta_E}}{Q} < R^{B_0-S-T+\eta_D+\eta_E}.$$

Thus the permissible budget is

$$\eta_D + \eta_E \leq \xi + S + T - B_0, \tag{3}$$

not merely ξ. The extra term comes from the actual least common multiple. Keeping it noticeably improves the source-restricted calculation. At this stage the relevant published input is the fourfold densely divisible estimate with

$$600\omega + 180\delta < 7. \tag{4}$$

Every improvement to the support still has to establish the hypotheses of that estimate [5].

Several apparently natural rearrangements fail. Splitting the fourfold requirement asymmetrically into owner orders $1 + 3$ or $3 + 1$ loses against the balanced $2 + 2$ choice in the first tests. Splitting each root into two fixed prime classes and asking all four groups to be singly dense is much worse: the group masses fluctuate too much. The resulting quotient is around 0.9925. These failures do not disprove more flexible group constructions, but they rule out the first versions.

The conditional prime-fragment law is useful because its moments are exact. Given a physical coordinate total t, the fragments have the scale-t Poisson–Dirichlet law with parameter one. The useful operation is to single out one fragment. The Palm identity removes a marked fragment u with weight du/u and leaves the same conditional law with total $t - u$. In particular,

$$\mathbb{E}(H_{>u} \mid t) = (t-u)_+,$$

and

$$\mathbb{E}(H_{>u}^2 \mid t) = \frac{t^2-u^2}{2}\mathbf{1}_{t>u} + \frac{(t-2u)_+^2}{2}, \tag{5}$$

where $H_{>u}$ is the total mass of fragments larger than u. Higher moments are sums over set partitions. Each block specifies one marked fragment, and after shifting every marked fragment by u, the remaining integral is an ordinary simplex monomial integral. This produces piecewise polynomial formulas with rational coefficients.

For example, expanding the exponential of the large-fragment mass gives the positive identity

$$\mathbb{E}\Big(e^{\lambda H_{>u}} \mid t\Big) = \sum_{n=0}^{\lfloor t/u \rfloor} \frac{1}{n!} \int_{\substack{v_j>u \\ \sum v_j \leq t}} \prod_{j=1}^{n} \frac{e^{\lambda v_j}-1}{v_j}\, dv_j.$$

There is no infinite-dimensional conditional integral left in this formula: it is a cumulative convolution exponential. This is why the fragment model is more tractable than it first appears. Yet an exact formula for a moment does not imply that a moment bound for a sharp event is sharp enough. The initial numerical exploration repeatedly separates those two issues.

That suggests polynomial majorants for the indicator in (1). Exponential majorants are easy to evaluate, but a single global Chernoff parameter is far too loose. The positive tail makes its expectation enormous. Binning by fragment size and radial mass helps substantially, but still leaves a factor close to two where the proof can afford only a small percentage. Bounded logistic and Gaussian majorants are much sharper numerically. Their Fourier representations reduce the problem to characteristic functions with explicit tails. However, the smallest eligible fragments introduce very high frequencies, so this approach exchanges one difficult error estimate for another.

There is also a finite-dimensional way to see the fragment law. Divide the primes into c fixed classes of equal harmonic density. Conditional on a coordinate total t, the class totals divided by t have distribution

$$(Z_1, \ldots, Z_c) \sim \mathrm{Dirichlet}(1/c, \ldots, 1/c), \qquad \mathbb{E}\prod_j Z_j^{n_j} = \frac{\prod_j (1/c)_{n_j}}{(\sum_j n_j)!}. \tag{6}$$

As c increases, the ranked class totals approach the prime-fragment law. The numerical behavior is encouraging, but a coarse cell treatment of all $41c$ variables incurs a boundary loss proportional to $41c$ times the mesh. It is more efficient to evaluate the limiting physical-coordinate integrals accurately, then use a strict margin to pass to a sufficiently fine fixed partition.

The exceptional configurations suggest a much simpler decomposition. Almost every offending fragment is the largest one. It is usually large on the normalized logarithmic scale, rather than close to the tiny source activation threshold. For the largest fragment u, the H_3 failure is just $3u > C$. Conditional on total t, the probability that every fragment is at most z is

$$\rho_D(t/z), \tag{7}$$

where ρ_D is the Dickman function. Thus the dominant restriction is separable across the physical coordinates. I can evaluate it by the same one-dimensional convolution machinery used for the original trial, replacing the coordinate density by its product with $\rho_D(t/z)$.

The many source tiers simplify further. Once redundant caps are removed, the chosen trial has only three outer radial regimes and three inner regimes. This makes the complete largest-fragment calculation manageable. Successively finer deterministic evaluations approach a quotient above one. But this restriction is only necessary for the full source condition: the smaller offending fragments remain to be controlled.

At first I divide by the completely unrestricted denominator I_0. That is unnecessarily costly. If I_* is the denominator on the full source support and I_H the denominator after imposing only the largest-fragment caps, then

$$I_* \leq I_H \leq I_0.$$

The middle quantity is already accessible to the same Dickman calculation. Using it recovers roughly 0.00053 in the quotient. This is a substantial change in the proof budget. A residual estimate that was previously too large may now fit.

The largest-fragment calculation is now manageable. I still have to cover every smaller offender.

The trial itself can be defined to be constant on rational physical-coordinate boxes. Then the finite sums are exact integrals of a legitimate step function. I do not need to pretend that a midpoint sum is an exact integral of the original smooth trial. The source tests use the appropriate upper box endpoints, and the frozen polynomial coefficients are replaced by exact decimal rationals. Smoothing will be a separate final approximation step.

The Dickman equation

$$\rho_D(v) = 1 \quad (0 \leq v \leq 1), \qquad v\rho_D'(v) = -\rho_D(v-1) \quad (v > 1)$$

allows validated cell integrals. First-order endpoint bounds are too wide after a large signed polynomial contraction. A validated trapezoid recurrence, with an explicit second-derivative remainder, makes the cell uncertainty small enough. The arithmetic then operates on intervals containing the exact finite sums. High precision controls numerical rounding; it does not replace the geometric inequalities that make the source-cell bounds valid [6].

There is a further way to avoid a misleading interval blowup. Suppose every true positive coordinate-cell mass is bounded above by $(1+\varepsilon)$ times a fixed reference mass. Since the integrands in the error estimate are nonnegative squares, I can first compare the product measures and then evaluate the square against the fixed reference measure. The resulting error factor is multiplicative in the number of coordinates. I should not instead expand the square into thousands of signed terms and charge each term the full independent cell uncertainty. That latter procedure destroys the cancellation without reflecting any actual uncertainty in the nonnegative integral.

This also explains why the final arithmetic checks have several layers. Exact combinatorial identities verify the marked angular moments and all coordinate-coincidence patterns. Directed numerical bounds enclose the finite convolutions and signed contractions. Separate support inequalities establish that the bins cover every possible source failure. None of these layers can substitute for another: an exactly computed integral over an incomplete exceptional set would still leave the proof unfinished.

The remaining source error has three parts. Small offending fragments are handled by a lower-tail estimate, the second-largest offender by a Palm calculation, and all higher ranks by factorial moments. This partition is exhaustive. An early sample contains no rank-three offender, but a larger sample eventually contains one. I must not turn the absence of an event in a sample into a support theorem.

For small fragments, take a bin $u \in [\ell, v]$ and retain the largest-fragment cap z. From (1), an offender in that bin requires

$$L_{\leq \ell} < s + 2v - C.$$

The exact positive one-coordinate kernel for a Laplace bound is

$$B_\lambda(t) = \exp_*\left(\mathbf{1}_{\ell < w \leq z} \frac{dw}{w}\right) * \left(e^{-\lambda t} \rho_D(t/\ell)\right), \qquad Q_\lambda = \left(\mathbf{1}_{[\ell,v]} \frac{dw}{w}\right) * B_\lambda. \tag{8}$$

Here $\exp_*$ denotes convolution exponential. The unmarked coordinates use B_λ and one marked coordinate uses Q_λ; the radial factor supplies the remaining exponential. This captures every offender rank in the small-fragment range. Unlike the original global exponential bound, it retains both the hard cap and the exact lower-fragment law.

There is a subtle cell issue. Several prime variables may lie in small logarithmic cells, and their sum can carry into the next physical-coordinate cell. Simply placing the whole mass in every possible target cell is safe, but wastes a factor of six or ten. Instead I can bound each prime density by a constant on its cell. Under that dominating product measure, the fractional positions are uniform, and the carry distribution is exact. If there are d such fractional variables, then

$$\operatorname{vol}\{j \leq U_1 + \cdots + U_d < j+1\} = \frac{\left\langle {d \atop j} \right\rangle}{d!}, \tag{9}$$

with Eulerian numbers in the numerator. Assigning these volumes, rather than duplicating entire cells, gives a sharp positive measure on the actual physical boxes. The signed polynomial can then be contracted against this fixed positive measure. Its cancellations are legitimate algebraic cancellations, rather than uncertain differences between unrelated interval endpoints.

For the next rank, I initially consider two ordered marks $p < q$ and the condition

$$q + 3p > C, \qquad q \leq C/3.$$

The exact two-mark density includes the requirement that no unlisted fragment exceed p. These are the Janossy factors. Dropping that requirement produces an extremely loose bound; a universal two-mark Markov estimate is several thousand times too large. The no-larger-fragment factors are doing real mathematical work.

There is a better one-dimensional identity. Condition on the largest fragment q. After the hard cap, rank two can fail only when

$$C/4 < q \leq C/3, \qquad \max(\text{remaining fragments}) > (C-q)/3. \tag{10}$$

Let A denote the Dickman density with cap q, and B the density with cap $(C-q)/3$. The exact conditional difference is a largest-fragment Palm integral of

$$A_{\text{owner}} \prod A - B_{\text{owner}} \prod B.$$

Call the coordinate containing the marked largest fragment its owner coordinate. There is a positive upper bound obtained by putting the difference either in that coordinate or in one of the other coordinates:

$$(A_{\text{owner}} - B_{\text{owner}}) \prod A + B_{\text{owner}} \sum_j (A_j - B_j) \prod_{l \neq j} A_l. \tag{11}$$

Only the largest-fragment variable is left to integrate. The physical owner must have total at least q, and the difference is actually supported where there is room for both large fragments. Enforcing these exact support facts removes a large artificial boundary loss.

For all ranks at least three above a cutoff p_0, the simple bound

$$\mathbf{1}_{N_{>p_0}\geq 3} \leq \binom{N_{>p_0}}{3} \tag{12}$$

suffices. Its expectation is the third factorial moment of the positive convolution measure. It includes all possibilities in which the marked fragments belong to the same physical coordinate or to different coordinates. A second-factorial shortcut for rank two looks adequate on a coarse grid, but worsens as the grid is refined and consumes almost the entire margin. The sharper largest-fragment difference is needed there.

The earlier source tiers require separate attention. Their owner condition is H_2, so the analogue of (1) is $L_{<u} < s + u - C$. A lower cutoff derived for rank-two offenders is not a lower cutoff for every rank. I use the same small-fragment argument with the correct offset, and the corresponding largest-fragment and factorial covers. The source activations also matter: the formally smallest activation may belong to a tier that never becomes active on the chosen root support. Using the actual active cores greatly reduces the tiny-fragment bounds.

The prime-face error is not a root probability. In the outer F^2 term of (2), every physical coordinate carries its usual profile weight. In the V^2/c term, the extra fiber on which the bad-root event is tested has no such profile factor. Its marked and unmarked kernels must therefore be treated separately. The inner error uses the squared face marginal directly. Keeping these distinctions is essential; small root-only probabilities do not by themselves certify a small prime-face loss.

There is another support issue in the earlier H_2 shells. Their source cap can be larger than the cap of the final outer radial piece. Squaring a sum of marginal pieces while discarding their different shared-row supports would be invalid. A positive decomposition into the appropriate nested shell squares gives a safe bound. Exact grouping by angular signature later makes the calculation shorter without changing the inequality.

The error budget has to be optimized at the level of these rigorous bounds. Four broad largest-fragment intervals are too expensive even though the underlying integral is small. Finer intervals, together with positive rational Young parameters chosen for each interval, reduce the total. A broad old bound can be replaced only when a complete finer cover of the same interval and radial shell has a smaller certified sum. This is a mathematical replacement of one upper bound by another; it must not omit the least convenient part of the interval.

The same phenomenon occurs in the lower-tail bounds. A broad source bin with one Laplace parameter is much worse than two narrower bins. Conversely, once enough margin has been recovered, a carefully optimized broad bin can be preferable because its entire bound still fits. Throughout, the criterion is the exact inequality

$$\rho \, \frac{J_H^- - E^+}{I_H^+} > 1, \tag{13}$$

with a positive numerator lower bound. The denominator replacement is useful only in that positive regime.

The completed forty-one-coordinate calculation records a quotient of approximately

$$1.0000609595724935.$$

At this point the source records a complete certificate for the diameter-188 tuple. The earlier numerical crossings, the necessary hard-cap inequality, and the root-only exceptional-set estimates were intermediate statements. They are not interchangeable with this final inequality.

One final analytic passage is needed. A finite band near zero must not be treated as one large rough fragment: its product can be large even though every constituent prime lies below an activation threshold. For each source tier I preseed every prime below its activation, exclude that seed from rough witnesses, and include its full mass in the smaller-prime prefix. Every activation is made a band boundary. The remaining positive bands are refined, and the virtual blocks are ordered by their actual logarithmic totals. An earliest-containing-block argument transfers their owner inequalities to actual primes. The probability of two relevant atoms in the same sufficiently narrow band tends to zero. After inward approximation of the hereditary support, bounded face operators give convergence in L^2. Thus a strict limiting inequality supplies some fixed finite source-valid partition and a smooth trial. It does not certify a particular finite number of prime colors merely because that number looked good numerically.

Can the same argument reach 186?

The combinatorial change is deceptively small. The exact finite searches give $H(40) = 186$ and $H(41) = 188$. The forty-one-point tuple just used has only one pair at distance greater than 186: its two endpoints. Every other pair is already close enough. Could I retain the forty-one-dimensional advantage and somehow remove only that endpoint pair?

A naive subtraction asks for an upper bound on simultaneous endpoint primality. The actual double-face mass of the frozen trial is about 0.0006003 in physical normalization. Even an optimistic elementary two-prime upper-sieve multiplier is about 38.9, giving a penalty around 0.0234. This is hundreds of times the certified surplus. The complete-prime mixed source leaves even less room for an additional upper-sieve factor. So the endpoint pair cannot be disposed of by a cheap auxiliary estimate.

Dropping an endpoint gives a legitimate forty-coordinate problem. Its unmasked optimum is already below one, around 0.997 after careful stabilization, and the source masks lower it further. Varying the inner radius, redistributing source budgets, adding more tiers, or increasing the nominal root while shortening the terminal support produces only small gains. The deficit is not a matter of finding one slightly better decimal parameter.

Numerical conditioning becomes a serious trap. Enlarged polynomial bases produce apparent quotients such as 1.06, 1.17, or even much larger values. Changing the grid or the small-eigenvalue cutoff destroys them. A small generalized eigenvector residual does not establish that the Gram matrix was computed accurately. I compare direct extended-precision convolution, orthogonal radial bases computed before contraction, and shellwise angular Gram normalization. These independent formulations agree on the stable values below one. Richer hinges, threshold indicators, inverse profiles, and angular products give modest improvements, not the dramatic crossings suggested by ill-conditioned power bases.

There is also a structural reason not to keep trying arbitrary scalar weights on larger tuples. Join two shifts when their difference is at most 186. This is an interval graph. If nonnegative coefficients c_i give a valid linear close-pair score, then the sum of the coefficients on every pairwise far-separated set is at most one. The weighted interval covering identity expresses such a score as a fractional combination of width-186 windows with total weight at most one. Hence

$$\sum_i c_i J_i(F) \leq \max_W \sum_{i \in W} J_i(F), \tag{14}$$

and every admissible window contains at most forty shifts.

The auxiliary coordinates do not rescue this. If y denotes the scored coordinates and z all the others, set

$$G(y) = \left(\int |F(y,z)|^2 \, dz \right)^{1/2}.$$

Tonelli gives $I(G) = I(F)$, and Minkowski increases the relevant one-face norms. Deleting coordinates preserves the hereditary source supports. Thus every such scalar linear construction reduces to a genuine trial in at most forty coordinates with no worse quotient. Larger diameter and more overlapping windows cannot evade this particular obstruction.

I next consider signed lifted support. Perhaps the full root can extend much farther out while every first marginal vanishes outside a smaller allowed face domain. The arithmetic identity behind this is valid: exact cancellation in the face marginal can occur before the distribution estimate is applied. But the marginal must vanish outside the complete buffered source domain, not just outside a radial simplex. Moreover, radial-only profiles and deletion-closed low-degree polynomial channels have a triangular Volterra structure; forcing their outer marginals to vanish forces the corresponding outer coefficients to vanish as well.

Threshold histograms provide genuinely more directions. A full k-coordinate configuration has $k+1$ possible counts above a threshold, while a face has only k. This leaves room for a null direction. The first coarse computation looks dramatic, but finer grids settle below one, around 0.996. Every combinatorially possible forbidden face has to be constrained, including faces with tiny probability. Dropping them on the basis of their numerical mass would create a false lift. Multiple thresholds are a richer possibility, but there is no successful source-valid crossing from this direction at this stage.

Matrix-valued scoring is more interesting because the scalar interval argument does not extend automatically. Let the Selberg root be a vector and attach a matrix Q_i to each shift. If

$$\sum_{i\in E} Q_i \preceq I$$

for every set E of mutually far-separated shifts, then the corresponding prime score is bounded by the denominator whenever there is no short pair. An exact rational example on the five-vertex path shows a genuine noncommutative gap: positive semidefinite shift matrices satisfy all independent-set inequalities, while every positive semidefinite window cover has cost at least 10/9. So the scalar reduction has found its limit.

That abstract gap is not yet a sieve construction. Assigning matrices to five isolated shifts contributes only five faces. They must be extended to every shift in an admissible host, and one common vector-valued trial must make the different face vectors align with the different matrix directions. A forty-three-point host containing the original tuple and two additional shifts is much better than a very large host, but the actual vector-valued variational searches still lose too much.

The rational path example is explicit enough to make the distinction concrete. Let P_0, P_1 be the two coordinate projections, let P_v project onto $(3, 4)$, let $P_{v^\perp}$ project onto $(4, -3)$, and let P_w project onto $(1, 3)$. The five matrices

$$\frac{5}{9}(P_0, P_1, P_w, P_v, P_{v^\perp})$$

satisfy all the independent-set constraints of the path. An exact dual certificate forces a window-cover cost at least 10/9. But there is no reason that the face covariances produced by a single common Selberg root should realize the abstract directions at no cost. Trials supported on different lower-dimensional coordinate sets are not a free substitute: their mixed normalizations and common source supports must also match.

The original endpoint graph illustrates this limitation. Complementary endpoint projections can score two orthogonal face directions fully. Making the actual endpoint faces nearly orthogonal, however, pushes the two vector components toward different forty-point cores. The lost variational mass is precisely what the abstract matrix problem did not measure. This is why I keep the pointwise matrix inequality and the analytic face optimization as separate questions.

Allowing indefinite Q_i requires two corrections. First, every independent subset must be checked; maximal independent sets suffice only when the matrices are positive semidefinite. Second, the usual mixed-square lower bound fails for a negative eigendirection. The identity

$$A^T Q_i A - (2A^T Q_i B - B^T Q_i B) = (A - B)^T Q_i (A - B)$$

is useful only with a nonnegative right-hand side.

There is a source-valid anisotropic version. Take one long root A and one short root C, and let

$$Q_i = \begin{pmatrix} a_i & b_i \\ b_i & c_i \end{pmatrix}, \qquad a_i \geq 0.$$

Then

$$a_i A^2 + 2b_i AC + c_i C^2 \geq a_i(2AB_i - B_i^2) + 2b_i AC + c_i C^2. \tag{15}$$

All terms on the right are long–short or short–short products, so the existing source rectangles suffice even when Q_i is indefinite. This is a real extension of the method, but the tested product families still do not cross one.

For the original endpoint pair, a singular matrix limit leads to a particularly transparent optimization. A small short component can carry large signed endpoint coefficients while contributing negligibly to the denominator. Optimizing that component is a Hilbert-space projection. For an endpoint-symmetric long trial, the unavoidable loss is

$$\frac{\rho}{I(A)} \int_{s(y)<T} \frac{D_T(y)^2}{T - s(y)}\, dy, \tag{16}$$

where D_T is the correctly truncated double marginal over the two endpoints and y comprises the other thirty-nine coordinates. The truncation matters: unrestricted and inner-restricted marginals are different operators. For the rich frozen trial, this loss is about 0.00828. Optimizing the entire fifty-six-dimensional symmetric feature span improves the resulting quotient only to about 0.9940, before source losses. A composite-companion formulation using only one-prime averages leads to the same projection penalty. These are several formulations of the same obstruction, rather than independent ways around it.

The remaining room may have to come from the arithmetic support.

A stronger distribution theorem would improve the scale, but its hypotheses must match the colored least-common-multiple moduli. Fixed-residue estimates with well-factorable weights do not automatically apply to these moduli. The improved smooth-modulus estimate in Stadlmann's work is especially tempting [4]. At the edge of its parameter range, the unmasked forty-coordinate quotient can move above one, though only by a small amount. A symmetric Baker–Irving minorant construction does not solve the problem: even its optimistic scale, combined with the standard variational upper bound, remains below one [3, 2].

There is a useful numerical scale check on the symmetric minorant idea. At the optimistic parameter endpoint appearing in this calculation, the scale is $\rho = 691/2636$. For a symmetric unit-simplex trial, the usual variational upper estimate gives

$$\rho M_{40} \leq \frac{691}{2636}\frac{40}{39}\log 40 < \frac{84993}{85670} < 1.$$

Even before accounting for the minorant's mass loss, that family cannot work. An asymmetric construction retains more room, but a minorant can take negative values on composites. Multiplying it by an indefinite mixed expression without a pointwise lower-bound argument would not preserve the desired inequality. This is the same sign issue that appeared in the matrix-valued experiment, in a different form.

The residue hypotheses are equally specific. A modulus in the multidimensional sieve collects primes assigned to many different shifts. Its residue is coherent, but generally it is not one fixed small integer independently of that assignment. A theorem with one or two fixed residue colors and well-factorable scalar coefficients cannot simply be inserted because its exponent is attractive. The exact CRT structure, the coefficient class, and every factorization step have to survive the proposed substitution.

The dense-modulus adaptation has a concrete technical issue. After dispersion, a companion factor is divided by a common factor q_0. Dense divisibility is not hereditary under this division.

There are exact squarefree examples that are strongly threefold, or even fourfold, densely divisible, while the relevant quotients are single primes too large to be singly dense at the original scale. This is not fixed merely by invoking the author's suggestion that an adaptation should be possible.

One apparent second obstruction is removable. The complete exponential estimate actually used comes from a squarefree-modulus bound; the stronger density assumption belongs to a different alternative estimate. Thus rough prime factors in that exponential modulus cause no new problem. The real issue is paying for the larger factor interval caused by q_0.

I can retain the loss from q_0 explicitly. With the enlarged factor parameter K and dispersion length H, the relations

$$K \le q_0, \qquad H \le H_0/q_0$$

give

$$\frac{K^3 H}{q_0^2} \le H_0, \qquad \frac{K^3 H^2}{q_0^2} \le H_0^2. \tag{17}$$

The resulting locally derived threefold source has parameter condition

$$240\omega + 80\delta < 3. \tag{18}$$

This is an extension established by the source argument under discussion, not a theorem to attribute verbatim to the published smooth-modulus statement. The common-residue and factorization steps have to remain in the proof.

Even granting this stronger source, the existing owner supports are too costly: their restrictions can consume several times the small unmasked surplus. A better exponent has to be paired with a cheaper sufficient condition for dense divisibility.

Which restriction is consuming the gain? Removing only the fourfold restrictions recovers about a thousandth in one common-draw comparison, while separately removing the earlier single and double restrictions changes the quotient by only a few ten-thousandths or less. The terminal owner geometry is the dominant loss. A refinement of a source tier is worthwhile only if it changes this geometry; adding many almost redundant early tiers will not supply the missing gain.

The first relaxation is almost embarrassingly simple. The two owner exponents need not be integers. If $r, s > 0$ and $r + s = 4$, the same proof multiplies an owner inequality of order r by one of order s and obtains the integer fourth-power insertion condition. The root masks become

$$H_{r+1,\xi}(D) \le a + \eta_D, \qquad H_{s+1,\xi}(E) \le b + \eta_E,$$

with the corresponding opposite-root maximum caps. Exact examples pass for fractional orders and fail for every integer split.

There is no need for the split to be constant in fragment size either. Let ϕ_D and ϕ_E be nondecreasing, with

$$\phi_D(u) + \phi_E(u) = 4u.$$

The owner masks use $L_D(u) + u + \phi_D(u)$ and the complementary expression. Monotonicity is exactly what is needed when the opposite owner contributes its first later prime. Piecewise-linear choices can pass examples for which every constant real split fails. Root-total-dependent exponents and direct upper envelopes for the strict tail give further hereditary variants.

The strictness can be checked with small exact integers. For example, take

$$Y = 30, \quad D = 7 \cdot 11 \cdot 19 \cdot 29 \cdot 31, \quad E = 2 \cdot 23 \cdot 37, \quad y_D = 3/5, \quad y_E = 71/5.$$

The split $r = 27/10$, $s = 13/10$ satisfies the owner, opposite-root, and gcd-budget conditions, while every integer split fails with the same budgets. Adaptive scales below one are permitted here because the conditions apply only to activated rough primes; replacing them silently by the usual unrestricted definition of single density would change the argument.

A second exact example separates size-dependent splitting from every constant real exponent. With $D = 7 \cdot 37 \cdot 47$ and $E = 11 \cdot 13 \cdot 17 \cdot 23$, one can use the multiplicative owner allowance

$$A(p) = \begin{cases} p, & p \le 40, \\ p^3/40^2, & p > 40, \end{cases} \qquad B(p) = p^4/A(p).$$

Both allowances are nondecreasing. The owner condition at one prime and the cross condition at another force contradictory bounds on any constant exponent, but this broken-line logarithmic allowance satisfies both. These examples establish a strict enlargement of the admissible support. They do not say that the enlargement lies where the variational trial puts useful mass.

These are genuine mathematical enlargements, but their numerical effect must be checked independently. An apparent gain from a fractional split reverses on a larger sample and another grid. The direct-tail implementation reproduces the original support exactly at its baseline parameters, which is a good consistency check, but the tested alternatives do not produce a robust improvement. Centered fragment features also help only slightly. For example,

$$C_q = \sum_i \left(\sum_{u \in D_i} u^q - \frac{t_i^q}{q} \right)$$

has conditional mean zero given the physical totals, and its exact conditional covariance makes it a convenient new variational direction. The first such directions improve the quotient by only a few parts in one hundred thousand.

An exact characterization of double dense divisibility provides a useful example of the slack in the standard insertion rule. For an already doubly dense integer N, the relevant insertion threshold can be written in terms of balanced singly dense divisors as

$$\theta_2(N) = Y \max_{\substack{d \mid N \\ d \text{ singly } Y\text{-dense}}} \min(d, N/d).$$

For $Y = 5$ and $N = 30$, this threshold is 25. The primes 13 and 17 can therefore be inserted even though they exceed $\sqrt{YN}$. This does not by itself supply the full Type-I factorization needed for the prime distribution theorem. Strong threefold density can still fail to provide a doubly dense factor with a singly dense complement at a prescribed large scale. Exact counterexamples occur throughout the relevant range of target proportions. I cannot weaken the recursive source condition merely because its simplest sufficient prime-chain test is inefficient [7].

The more promising observation comes from examining what strong dense divisibility really requires. The familiar sufficient insertion condition $p^r \le YN$ can be far from necessary. Suppose first that S is Y-smooth and $p > Y$ is the only rough prime. Then

$$pS \text{ is strongly } r\text{-fold } Y\text{-dense} \quad \Longleftrightarrow \quad S = A_1 \cdots A_r \text{ with every } A_j \ge p/Y. \tag{19}$$

For necessity, use the recursive split of orders $(r-1, 0)$ at targets tending to p from below. The selected divisor cannot contain p, so it consumes a smooth factor of size at least p/Y. Repeat on the remaining factor. For sufficiency, reserve one of the A_j as a bridge between the two possible placements of p in an arbitrary recursive split. The bridge is large enough that the two target ranges overlap.

The same protected smooth factors support arbitrarily many rough primes, provided every rough prime is at most $Y \min A_j$. But this is still not useful numerically in the late source tiers. There is too little genuinely smooth seed mass. Even a generous necessary mass test rejects nearly every relevant configuration. Designing the primal sieve coefficients directly on such a nonhereditary reservoir set does not help if the set itself carries almost no mass.

The decisive strengthening is to let the protected factors themselves contain rough primes. The resulting theorem is:

$$\begin{gathered} Q \text{ squarefree}, \qquad A_1, \ldots, A_r \mid Q \text{ pairwise coprime and singly } Y\text{-dense}, \\ P^+(Q) \le Y \min_j A_j \quad \Longrightarrow \quad Q \text{ strongly } r\text{-fold } Y\text{-dense}. \end{gathered} \tag{20}$$

There may be arbitrary leftover primes in Q. This is substantially stronger than the smooth-reservoir statement.

The proof uses induction on both the order and the number of primes. Remove the largest prime p. If it belongs to a protected group, the singly dense prime-prefix condition shows that the remaining part of that group is still singly dense and has size at least p/Y. The smaller integer therefore retains the required protected groups.

For an extreme recursive orientation, take $r-1$ intact protected groups on one side. The omitted protected group, after removing p if necessary, is large enough to absorb all the leftover primes by the ordinary single-density insertion rule. Its divisor chain bridges the low divisors of Q/p to the high divisors containing p. For an interior orientation $j+k=r-1$, assign j protected groups to one side, k to the other, and keep one bridge group. Lower-order induction supplies the small-target and large-target factorizations. If there is a gap between those ranges, the product of the k reserved groups itself lies in the required target interval. This covers every orientation, not just the balanced one.

Exact examples show why this matters. Four singly dense groups can certify strong fourfold density even though the old fourth-power prime insertion test fails at more than one rough prime. On the fragment configurations, constructive partitions into rough singly dense groups succeed on a substantial fraction of cases where smooth reservoirs never succeed. The remaining problem is to turn that existential partition into a hereditary, separately imposed condition on the two Selberg roots.

A protected group $A \mid D$ can be made singly dense whenever D lies outside its core by imposing

$$H_{2,\xi}(A) + s_{D/A} \le a + \xi. \tag{21}$$

Indeed, $s_D > a$ turns this into the ordinary single-density prime-prefix inequality inside A. Both terms on the left decrease under deletion. A companion bound on $s_{D/A}$ ensures that the group is large enough to protect the largest prime. Fixed disjoint coordinate blocks supply coprime groups, and using merged groups $[D_j, E_j]$ gives larger protected factors. The least common multiple of two singly dense squarefree integers is singly dense, because every prime can use its original owner's smaller-prime prefix.

Hereditary support is the constraint that makes these constructions delicate. A statement saying that the current integer contains four good groups is not enough. The sieve transformation can expose divisors of that integer, and those divisors may have lost the small primes that made a group dense. The safe core in (21) is what repairs this: if deletion destroys too much protected mass, the remaining root falls back into a region where no special source condition is required. If it remains outside the core, the same monotone inequality still proves the necessary density. This is why a superficially stronger partition test can be less useful than a weaker condition with the right monotonicity.

For merged groups, the fixed arithmetic partition is also doing more than simplifying notation. A large prime cannot be assigned independently to two different physical shifts after the usual presieving, so disjoint coordinate or prime-residue classes give genuinely coprime protected groups. Within one class, however, D_j and E_j may share arbitrary primes. The proof must retain those common factors until the least common multiple is formed. Throwing them away early would turn a helpful cancellation in (22) into an unnecessary loss, or worse, would assume a density property for a quotient that need not have it.

This is particularly important when trying several colorings. A union is safe if each possible outer witness works with every permitted inner root. It is not automatically safe to take independent unions on both sides: one outer root might be accepted by the first partition and one inner root by the second, with no common partition certifying their least common multiple. The common inner cap in the one-root construction is a way of enforcing that compatibility. Any later relaxation has to supply an equally explicit argument for the cross pairs.

The numerical contrast is striking. In the late tiers, the smooth-reservoir necessity test essentially never passes. Allowing rough singly dense protected groups makes a constructive partition succeed on about half the sampled configurations in the initial screen. That screen only establishes that an individual configuration has a witness. It does not yet establish compatible witnesses for every pair of independently selected roots. The fixed-block and rectangular constructions are attempts to bridge exactly that gap.

The first fixed-block implementations lose too much. With the improved threefold source, only three protected groups are needed, but forcing each predetermined group to work still costs one or two percent. Prime-residue classes preserve coordinate symmetry and provide a natural alternative to physical-coordinate blocks. Their conditional masses have the exact Dirichlet law from (6), and different fixed residue partitions can be combined using the Chinese remainder theorem. A finite union of hereditary outer supports remains hereditary, provided every accepted outer choice is compatible with the same inner restriction.

There is also a better way to account for the common divisor. Let D_j, E_j be a fixed class, write $o_{D,j} = s_D - s_{D_j}$ and similarly for E, and put $L_j = [D_j, E_j]$. If g and g_j are the normalized global and classwise gcd masses, then

$$s_{L_j} = s_Q + (g - g_j) - o_{D,j} - o_{E,j} > B_0 - o_{D,j} - o_{E,j}. \tag{22}$$

The gcd term helps rather than hurts. Thus the protected-size condition follows from

$$o_{D,j} + o_{E,j} + \max(M_D, M_E) \le B_0 + \xi,$$

without a fixed global maximum cap or a separately subtracted worst-case gcd allowance. This condition has a hereditary rectangular implementation: choose constants d_j, e_j and require

$$\begin{aligned} &o_{D,j} + M_D \le d_j, &&o_{D,j} \le e_j,\\ &o_{E,j} \le B_0 + \xi - d_j, &&o_{E,j} + M_E \le B_0 + \xi - e_j. \end{aligned} \tag{23}$$

Together with (21), this gives a source-valid cap-free merged-group construction.

The protected-group theorem asks the merged group $L_j = [D_j, E_j]$ to be singly dense. Why am I still asking both D_j and E_j to be singly dense separately? Their prime factors may fill one another's gaps. I need a joint prime-prefix inequality which can still be enforced by separate hereditary conditions on the two roots.

Keep the logarithmic normalization $s(D) = \log_R D$ and $u = \log_R p$, and define the inclusive tail of a fragment collection G by

$$T_G(u) = \sum_{v \in G,\ v \ge u} v.$$

The usual obstruction to single dense divisibility is $T_G(u) + u$. If the group has total size $s(G)$, then

$$T_G(u) + u \le s(G) + \xi \quad \Longleftrightarrow \quad u \le \xi + \sum_{v \in G,\ v < u} v.$$

This is exactly the logarithmic form of the familiar prime-insertion condition: the next prime should not be larger than the density scale times the product of the preceding primes.

Suppose that the factors of D and E have been partitioned into corresponding groups D_j, E_j, and that $L_j = [D_j, E_j]$. I first imposed the single-density condition on both D_j and E_j. That was much stronger than needed. A better quantity was

$$K_\alpha(G;m) = \sup_{\xi<u\le m} (T_G(u) + \alpha u), \qquad 0 \le \alpha \le 1.$$

The cap m is fixed in advance, and bounds every relevant prime fragment. Since the tail is constant between consecutive fragments,

$$K_\alpha(G;m) = \max\left\{\alpha m, \max_{u\in G,\ u>\xi}\left(\sum_{v\in G,\ v>u} v + (1+\alpha)u\right)\right\}.$$

The endpoint term αm is essential. A prime may occur in the opposite root without occurring in this group, so the envelope has to control every possible threshold, not just the primes presently visible in G.

Let B be the lower endpoint of the modulus band, and put $o_D = s(D)-s(D_j)$, $o_E = s(E)-s(E_j)$. The paired conditions

$$o_D + K_\alpha(D_j;m) \le d_j, \qquad o_E + K_{1-\alpha}(E_j;m) \le B + \xi - d_j$$

have a useful consequence whenever $s([D,E]) > B$. At a prime fragment u belonging to L_j, their sum gives

$$o_D + o_E + T_{D_j}(u) + T_{E_j}(u) + u \le B + \xi.$$

The shared primes have been counted twice on the left. This is harmless, and in fact supplies the correct direction of the gcd correction. Subtracting the tails from the total sizes shows that the prefix of L_j below u exceeds $u - \xi$. Thus L_j is singly densely divisible even when one of its two contributing groups is not.

I should check the gcd calculation explicitly, because an apparently innocent cancellation here would affect the whole source theorem. Let g be the total logarithmic mass of $\gcd(D,E)$, and let $g_{<u}$ be the common mass in the selected group below u. The paired inequality gives

$$\mathrm{prefix}_{D_j}(u) + \mathrm{prefix}_{E_j}(u) \ge s([D,E]) + g - B - \xi + u > g - \xi + u.$$

The prefix in the least common multiple is the sum of the two prefixes minus $g_{<u}$. Since $g_{<u} \le g$, the required inequality follows in the correct direction. In particular, no coprimality assumption has slipped into the argument. The size calculation has the same feature: if g_j is the common mass in the whole selected group, then

$$s(L_j) = s([D,E]) + g - g_j - o_D - o_E > B - o_D - o_E.$$

The common factors help both estimates rather than creating a loss.

The same inequality gives a second benefit. Evaluating the two envelopes at m gives

$$o_D + o_E + m \le B + \xi.$$

Since the actual least common multiple is above B, the merged group has size greater than $m - \xi$. Thus the same conditions provide both the dense-divisibility property and the size required of a protected reservoir. I do not need a separate family of size restrictions. With $\alpha = 1/2$, the per-root obstruction has order $3/2$, rather than order 2. This explains why sharing the burden between the roots could be substantially cheaper.

A small integer example made the distinction concrete. Take $Y = 5$, $D = 2310$, and $E = 13$. Their least common multiple is 30030. The three disjoint groups

$$2002 = 2\cdot 7\cdot 11\cdot 13, \qquad 3, \qquad 5$$

are singly 5-densely divisible, and their sizes are large enough relative to the largest prime 13. The protected-group lemma therefore supplies the higher-order divisibility property. On the other hand, the older cubic insertion condition already fails at 7, since

$$7^3 = 343 > 5(2\cdot 3\cdot 5) = 150.$$

The inner root 13 is not itself singly 5-densely divisible. So this really does exhibit a gain from merging complementary factors, rather than a different notation for the old condition. It is also only an arithmetic example. A strictly larger permitted set of moduli does not automatically improve the final variational quotient enough; the trial must put useful mass in the newly permitted region.

I also tried making the partition adaptive. For example, two protected groups could come from D, while the whole of E supplied the third. Initially I charged the largest possible gcd to each group. The actual modulus band gives a much better bound. If $s(D) \leq S$, $s(E) \leq T$, $a = B - T$, and $g = s(\gcd(D, E))$, then

$$g = s(D) + s(E) - s([D, E]) < s(D) - a.$$

The excess of D beyond its safe core already pays for removing its common factors with E. This is the right way to use the modulus condition; a uniform worst-case gcd penalty throws away this cancellation.

The adaptive partition problem itself was finite. Given fragments in increasing order, I could record all attainable loads of the first of two groups, and discard a transition whenever inserting the next fragment violated the prefix condition. A greedy rule was not enough. It could put two small fragments in different groups and leave neither group large enough to receive a later fragment, even though putting those small fragments together would have worked. There was also a distinction between finding one acceptable partition and describing the support of the sieve coefficients. The existential support, consisting of all roots admitting some acceptable partition, has a useful divisor-hereditary interpretation. An arbitrary greedy selection need not have that property. A trial can still be supported on a smaller selected set, provided its divisor closure lies inside the analytically permitted support. I needed to keep these two assertions separate.

These reservoir constructions clarified the geometry, but they did not immediately produce the strongest numerical inequality. The more decisive simplification was to divide the owner order itself. For a third-order insertion condition, there is no need to assign integer orders 2 and 1 to the two roots. One can use functions ϕ_D, ϕ_E satisfying

$$\phi_D(u) + \phi_E(u) = 3u.$$

The root conditions control an inclusive tail plus the appropriate owner function, together with the cross-cap conditions needed when a prime occurs only in the opposite root. Adding the two estimates then recovers the order-three condition for the actual least common multiple. The balanced choice $\phi_D = \phi_E = 3u/2$ was a natural first test.

The order split is useful only if its support remains hereditary. With $a = B - T$, $b = B - S$, and $\gamma = S + T - B$, the two owner ceilings could be allocated so that their extra allowances satisfied

$$\eta_D + \eta_E \leq \xi + \gamma.$$

Since $a + b = B - \gamma$, their sum is at most $B + \xi$, which is exactly the budget needed after using $s([D, E]) > B$. Deleting prime factors decreases the inclusive tails and the fixed cross-cap obstructions. This gives the hereditary support needed by the inverse Selberg transformation. It does not say that arbitrary divisors of a densely divisible integer preserve the same density parameter. That latter statement is false in the form one would need, and was the reason for retaining the explicit gcd loss in the distribution argument.

Here the distinction between a published input and a new argument matters most. The available results of Polymath and Stadlmann supplied the distribution machinery and the lower-order estimates, but the strong third-order extension being used here was part of the argument under development. I could not simply relabel it as an existing theorem. Its treatment of the dispersion gcd, coherent residue classes, and the successive factorization had to remain in the proof. The finite arithmetic tests only checked consequences and examples of those claims; they did not establish the asymptotic distribution estimate [5, 4].

Must the owner functions be balanced? The outer and inner roots play different roles in the variational problem, so there is no reason to insist on the same linear penalty. I can take

$$\phi_D(u) = \min\left(\frac{3}{2}u, L\right), \qquad \phi_E(u) = 3u - \phi_D(u).$$

The outer penalty stops increasing at L, while the inner one becomes steeper after the breakpoint $2L/3$. Their sum remains exactly $3u$. This is not an improvement for every root in both positions. It is a way to spend the available source strength where the chosen trial benefits most from it.

The useful plateau is eventually $L = (23/40)C$, where C is the inner owner ceiling. The earlier choice $3C/5$ already suggested that this asymmetry could help. The new support gives enough room to return to the minorant comparison, provided I preserve the signs in the mixed form.

I want to express the gain as a first-form inequality. Let J_0 denote the base prime-face form, J_1 the enlarged form accessible through the minorant, and J_f the unrestricted comparison form. These use one common outer trial and nested inner face spaces, so $0 \le J_0 \le J_1 \le J_f$. Put

$$\Delta = J_1 - J_0, \qquad E = J_f - J_0.$$

The earlier sign objection still has to be resolved. Write the minorant as $\mu = P - \mathfrak{b}$, where P is the prime indicator and $\mathfrak{b} \ge 0$ is supported on the exceptional composites. The common prime-factor cap lies below the least prime factor of those composites. Thus the distinguished divisor is one both on genuine primes and on the exceptional support, and the same face identity applies in both places.

Let A be the outer Selberg expression, let B_0 be its base inner completion, and put $X = A - B_0$. If C is an additional inner completion, then for every $\eta > 0$,

$$PX^2 \ge 2\mu XC - PC^2 - \mathfrak{b}(\eta X^2 + \eta^{-1}C^2).$$

Indeed, the difference between the two sides is exactly

$$P(X - C)^2 + \mathfrak{b}\big(\sqrt{\eta}X + C/\sqrt{\eta}\big)^2 \ge 0.$$

No sign assumption on μ is needed. The extra negative terms pay explicitly for the exceptional composites, so the final witness can still count genuine primes.

In the normalized averaged forms, the two exceptional square estimates are bounded by KE and $c\Delta$. Here $c = 1 - m$ is the minorant deficiency, and K is the weighted exceptional-composite comparison constant. The support analysis supplies that comparison; the pointwise identity alone does not determine K. Optimizing η bounds the exceptional cross loss by $2\sqrt{cKE\Delta}$. Scaling the additional completion by $t \ge 0$ then gives the extra score

$$2t(m\Delta - \sqrt{cKE\Delta}) - t^2\Delta.$$

Its maximum is $(m\sqrt{\Delta} - \sqrt{cKE})_+^2$, with value zero when no positive increment is available. The resulting gain is therefore

$$J_0 + \big(m\sqrt{\Delta} - \sqrt{cKE}\big)_+^2.$$

The square root term makes direct optimization less convenient. For $0 < \lambda < m$, Young's inequality gives a linear lower bound

$$J_\lambda = J_0 + a\Delta + bE = d_0 J_0 + aJ_1 + bJ_f,$$

where

$$a = m^2 - m\lambda, \qquad b = (1 - m/\lambda)cK, \qquad d_0 = 1 - a - b.$$

The sign of b is negative. I need to preserve it throughout the argument. Treating all three terms as positive source forms would destroy the pointwise minorant comparison.

I should also check that the positive-part notation is compatible with this linear lower bound. When $m\sqrt{\Delta} \ge \sqrt{cKE}$, it is just Young's inequality applied to the cross term. In the other case,

$cKE > m^2\Delta$. Because $b < 0$, substituting that lower bound into the linear expression makes its increment at most

$$\left(m^2 - m\lambda + (1 - m/\lambda)m^2\right)\Delta = -\frac{m}{\lambda}(m-\lambda)^2\Delta \le 0.$$

Thus the linear expression is below J_0, as it must be when the positive-part gain vanishes. The inequality is global; it does not depend on choosing a candidate for which the square root happens to be positive in a floating-point calculation.

The fixed rational choices were

$$m = \frac{49999}{50000}, \qquad K = \frac{17}{50}, \qquad \lambda = \frac{1}{125}.$$

They give

$$a = \frac{2479900401}{2500000000}, \qquad b = -\frac{843183}{1000000000}, \qquad d_0 = \frac{44415113}{5000000000}.$$

Stadlmann's refinement, Section 4.8, supplies the sharper deficiency bound $103/10^7$, and I checked what it would save [4]. However, the final fixed choice retained the more conservative $c = 1/50000$. The gain had to survive a complete argument with that value, not an optimistic replacement inferred from numerical integration [3].

The trial was a positive one-coordinate profile times a rational symmetric polynomial. Schematically it had the form

$$F(\mathbf{t}) = \prod_{i=1}^{40} g(t_i) \sum_{\nu,r} c_{\nu,r} \left(\sum_i t_i - c_0\right)^r \prod_{a\in\nu} \left(\sum_i t_i^a\right),$$

with a fixed rational center c_0 and 77 rational coefficients. The profile parameters were

$$g(t) = \frac{w}{1+\alpha t} + \frac{1-w}{1+\beta t}, \qquad (w, \alpha, \beta) = \left(\frac{21}{200}, \frac{1}{100}, \frac{907}{5}\right).$$

Writing the trial this way made the low-dimensional symmetric structure explicit. It did not reduce the source condition to a condition on the total $\sum t_i$. Each physical coordinate still contained its complete collection of prime fragments.

This distinction explained an initially misleading observation. Sampling the full source masks sometimes found no failures for a long stretch. It was tempting to conclude that the largest-fragment caps already implied the other restrictions. They did not. A root can contain two or more moderately large fragments whose combined tail violates an owner condition, while each individual fragment respects the largest-prime cap. Rare failures can have a disproportionately large weight under a signed, highly concentrated trial. A sample with no failures therefore gave no proof that the loss was zero.

Importance sampling helped identify the troublesome regions. I looked at configurations with two marked fragments, either in different physical coordinates or in the same coordinate. The correction factor had to include all possible choices of the marked owners, and the ordered two-mark law could not be replaced by an unordered one without changing its normalization. These calculations were useful diagnostics. Their real purpose was to suggest a positive integral majorant that could subsequently be enclosed without statistical uncertainty.

The same caution applied to the main variational calculation. A midpoint approximation could show a positive surplus while the corresponding inward full-cell calculation was negative. In dimension 40, a small displacement in each coordinate becomes a substantial displacement of the total. A cell that straddles the boundary of the admissible region cannot be treated as if its midpoint represented the entire cell. The positive midpoint value was evidence about a candidate; it was not yet the desired inequality.

I therefore fix the rational trial first, and use the same physical grid for its denominator, its cap-only numerator, and every source correction. A coarse source calculation could guide the choice of bins or the choice of a candidate. It could not be inserted into a finer-grid main calculation merely because both looked close to a common limit. The signed polynomial has cancellations, and there was no proved monotonicity that would justify such a substitution.

The fragment measure also needs care. Before imposing a prime cap, its total-coordinate push-forward is Lebesgue measure. After imposing a largest-fragment cap ζ, it is

$$\rho_{\mathrm{D}}(t/\zeta)\,\mathrm{d}t,$$

where ρ_{D} is the Dickman function. Some relevant totals exceed ζ, so this factor cannot be replaced by 1 everywhere. The one-coordinate law, its marked versions, and the face laws must all use the same capped measure. The appearance of fragment probabilities here is a consequence of the finite-band Selberg limit, not an assumption that primality events are independent.

The order of limits matters here as well. The prime fragments are first placed in finitely many positive logarithmic bands, with strict source margins fixed. The asymptotic sieve calculation is performed for that fixed data. Only afterward are the bands refined and the trial approximated from the interior. Collisions of two fragments in a single coordinate band have a bound involving

$$\sum_I \left(\int_I \frac{\mathrm{d}u}{u}\right)^2,$$

which tends to zero on a compact positive range as the bands are refined. The small-fragment tail is handled separately. This gives a controlled passage to the fragment model. It would be quite different to assume an independent random model for the primes and then read its predictions as a theorem.

There is a useful elementary check on the size of the face operator. If E_i integrates out the i-th coordinate, put

$$w_i = S - s_{\hat{i}} + (k-1)t_i.$$

Weighted Cauchy–Schwarz, together with $0 \le \rho_{\mathrm{D}} \le 1$, gives

$$\int \frac{\mathrm{d}\nu(X_i)}{w_i} \le \int_0^{S-s_{\hat{i}}} \frac{\mathrm{d}t_i}{S - s_{\hat{i}} + (k-1)t_i} = \frac{\log k}{k-1}.$$

Since $\sum_i w_i = kS$, it follows that

$$\sum_i E_i^* E_i \le \frac{Sk\log k}{k-1}.$$

For $k = 40$ and $S = 2742997/2624989$, this is less than 4. Even that last numerical comparison can be made rational: the first twelve terms of the exponential series at $369/100$ already exceed 40, so $\log 40 < 369/100$. This gives an explicit justification for the constant 4.

An apparently stronger estimate had to be discarded. If P projects onto source-good outer roots, one might hope that

$$\|(1-P)Af\|^2 \le C^2 \|(1-P)f\|^2$$

follows from $\|A\| \le C$. It does not: P and A need not commute. The operator can move mass from a good input region into a bad output region. The correct restoration needed actual marked face integrals, or a valid Cauchy–Schwarz bound for those integrals. This was precisely the kind of error that a plausible small numerical residual could conceal.

The first-form restoration is easier to organize. Let A denote the cap-only signed hybrid, and B the hybrid with the genuine inner source projections. The coefficients d_0, a are positive and the negative full-face term is unchanged, so $A - B$ is positive. Its loss can be bounded by a positive quantity β. Removing the bad outer roots introduces a second positive upper bound $\mathcal{E}_O$. I then need

$$\rho_*\left(J_{\lambda,\mathrm{cap}} - \mathcal{E}_O - \beta\right) > I_{\mathrm{cap}}.$$

The denominator becomes smaller when the outer support is restricted, which is favorable. I need complete upper bounds for the two numerator losses.

The expansion behind the outer correction was worth keeping in view. If $e = (1 - P)f$, then

$$\langle Pf, BPf\rangle = \langle f, Bf\rangle - 2\,\mathrm{Re}\langle e, Bf\rangle + \langle e, Be\rangle.$$

The two positive source terms in B remain positive, while the negative full-face term is bounded below using the face operator norm. Thus the last term has a controlled negative part. For the cross term, a correct estimate involves the mass of the actual bad-output event, rather than a fictitious commutation of P and B.

On each positive source component I could instead apply

$$2|UV| \le \eta|U|^2 + \eta^{-1}|V|^2, \qquad \eta > 0,$$

to the root value and the corresponding face value. This produces two positive square integrals. The parameter η can be chosen rationally after their approximate sizes are understood, and the final upper bound uses their outward enclosures. This explains both the usefulness and the cost of the source decomposition: it allows different regions to use appropriate bounds, but every region must still be integrated with the same exact trial and physical measure.

The nonlinear owner functions did not force an unwieldy collection of separate bad events. For the outer root, $\phi_D(u) \le 3u/2$, so an outer failure is covered by the balanced $H_{5/2}$ obstruction. For an inner fragment below the kink, the same description is exact. Above the kink, a nonlargest fragment has a companion at least as large. Hence its balanced obstruction is already at least

$$2u + \frac{3}{2}u = \frac{7}{2}u > \frac{7}{3}L.$$

The rational source parameters make this exceed the relevant ceiling. The high nonlinear branch is therefore covered as well. Together with the H_2 failures in the lower-order source tiers, this reduced the correction to six roles: two outer roles, two base-inner roles, and two enlarged-inner roles.

There was still an issue at the very small activation thresholds. Some formally present source tiers could never be reached by the actual least common multiple of two roots in the inward full-cell support. Such a tier could be removed by an exact support comparison. This was preferable to discarding a thin radial strip simply to avoid a difficult integral. Where a tier remained, however, its entire activated interval had to be covered, including its smallest fragments.

For one of the six roles, write the obstruction in the form

$$H_m(p) = \sum_{q>p} q + mp > U, \qquad m \in \left\{2, \frac{5}{2}\right\}.$$

The largest-fragment cap z excludes a failure caused by the largest fragment alone. I split the possible witnesses at a positive threshold w. A low witness lies in an interval $(l, u] \subseteq (\xi, w]$. Since its inclusive tail is at most the total s, it also forces

$$s > U - (m-1)u.$$

It must lie beyond the core of whichever source tier is active. Thus, among tiers whose activation is below u, the appropriate radial lower bound is obtained from

$$\min_{\text{eligible tiers } t} \max\{\mathrm{core}_t, U_t - (m-1)u\}.$$

This is stronger than minimizing the two quantities separately. It also makes clear why the activation and the radial core must remain coupled to the same tier.

For the low witness I used a positive Chernoff majorant. If $L_{<l}$ is the total mass of fragments below l, then the failure implies

$$s - L_{<l} + (m-1)u > U.$$

For any $\theta > 0$, the indicator is bounded by

$$\exp(\theta(s - L_{<l} + (m-1)u - U))\,.$$

The witness itself is marked with the logarithmic prime measure $\mathrm{d}p/p$. Summing over all possible witnesses only overcounts the event, which is acceptable for an upper bound.

There was a further counting refinement. If $z + mw \le U$, a low failure cannot be supported by only two fragments at least as large as its witness. Indeed, those two would give at most $z + mp \le U$. There must be at least three. If D counts designated witnesses and N counts the fragments of size at least l, then

$$D\binom{N-1}{2}$$

is another positive majorant. This led to marked factorial moments with one witness and two companions. It was a useful refinement, but it did not by itself close the estimate. In some of the dominant bins it improved very little, so I still had to refine the geometric partition and the radial information.

For high witnesses, configurations with three or more fragments above w are covered by $\binom{N_w}{3}$. The remaining case has two large fragments, say $q \ge p > w$, and

$$q + mp > U.$$

Consequently

$$q > \frac{U}{m+1}, \qquad p > \frac{U-q}{m}.$$

I could integrate over the largest fragment q, and use a positive difference of Dickman laws to insist on a companion above the required threshold. The companion may lie in the same physical coordinate as q, or in another coordinate. On a prime face, either of those marked coordinates may also be the one integrated out. All those placements must be included.

A compact algebra kept track of these placements. For one physical coordinate, let P be the ordinary law, A the largest-owner law, B the distinct companion law, and C the law in which both marks share that coordinate. In the positive ring with $x^2 = y^2 = 0$, write

$$P + xA + yB + xyC.$$

The coefficient of xy in the product over coordinates is

$$\sum_i C_i \prod_{j\ne i} P_j + \sum_{i\ne j} A_i B_j \prod_{r\ne i,j} P_r.$$

It includes both same-coordinate and different-coordinate marks, with the correct ordered owner distinction. If a coordinate e is erased to form a prime face, the four possibilities are

$$C_e P_{\mathrm{ret}}, \qquad A_e y_{\mathrm{ret}}, \qquad B_e x_{\mathrm{ret}}, \qquad P_e (xy)_{\mathrm{ret}}.$$

Here the retained symbols denote the corresponding coefficients of the retained-coordinate product. Omitting one of these terms would omit a genuine physical placement of the source failure. Conversely, their positivity makes overcounting transparent and safe. This was preferable to trying to infer the erased-coordinate contribution from a scalar probability of having two large fragments somewhere in the root.

This gave a complete low-witness, two-large-fragment, and higher-rank cover. The pieces can overlap, because they are positive upper bounds. A gap between two pieces is fatal. In particular, a lower bound for the second-largest fragment that is valid only in the two-large-fragment case cannot be used to remove the low-witness part of the all-rank cover.

One later improvement came from a very simple physical restriction that the first majorants had ignored. Fix a largest-prime bin $(q_-, q_+]$, and let

$$L_q = \log \frac{q_+}{q_-}.$$

If a coordinate cell has upper endpoint t^+, it can own q only when $q < t^+$. Define the clipped logarithmic mass

$$\mathscr{L}(v) = \begin{cases} 0, & v \le q_-, \\ \log(v/q_-), & q_- < v < q_+, \\ L_q, & v \ge q_+. \end{cases}$$

The available fraction of the original bin mass is at most

$$\lambda_{\mathrm{own}}(t^+) = \frac{\mathscr{L}(t^+)}{L_q}.$$

If the companion p is in the same coordinate, then

$$t^+ > q + p > q + \frac{U - q}{m},$$

which gives the stronger fraction

$$\lambda_{\mathrm{same}}(t^+) = \frac{1}{L_q} \mathscr{L}\left(\frac{mt^+ - U}{m-1} \right).$$

Every term of the positive marked expansion has exactly one largest-prime owner. I could therefore insert the appropriate fraction once, without changing the unmarked laws or charging the external mass L_q twice. The other envelopes had already been maximized over the entire bin, and remained valid on the smaller feasible part. This was a real tightening of the majorant, not a new assumption about the event.

I also considered replacing a union bound for companions by an exact telescoping product difference. That led to another useful warning about interval arithmetic. An upper bound for A minus an upper bound for $A - B$ is not a lower bound for B. Subtracting two separately rounded objects in that way can reverse the inequality. A valid telescoping expression needs compatible positive laws and a genuinely lower-enclosed factor. The feasible-owner fractions were simpler to retain: they directly multiply existing positive majorants and preserve every owner placement.

The remaining calculation was built from positive convolutions. For a fixed prime band, the cell weights for unmarked fragments and for one or more distinguished fragments are nonnegative. Exponential generating functions account for unordered collections of fragments, and factorial marks account for the required companions. Full-cell carries are included when several fragments occupy cells whose lower endpoints add to a given index. The symmetric polynomial trial reduces these integrals to a finite collection of moment signatures. This is what makes a forty-coordinate integral manageable without replacing its source conditions by a one-dimensional surrogate.

The sign of the trial coefficients still mattered. It would be safe but far too expensive to replace every coefficient by its absolute value. Instead, the positive moment measures were enclosed first, and coefficients belonging to the same moment signature were combined before the signed contraction. The algebra had to describe one coherent measure throughout. It would not be legitimate to select unrelated upper envelopes for different signed pieces and then claim that their cancellation represented the original square.

Exact positive polynomial arithmetic offered a particularly clear way to handle the convolution. With a fixed denominator $Q = 2^b$, store a nonnegative polynomial between two integer polynomials divided by Q. If $A^{\pm}, B^{\pm}$ are the integer coefficient bounds, multiplication uses

$$\frac{1}{Q}\left\lfloor \frac{A^- B^-}{Q} \right\rfloor \le AB \le \frac{1}{Q}\left\lceil \frac{A^+ B^+}{Q} \right\rceil,$$

with floor and ceiling taken coefficient by coefficient. Arbitrary-precision interval arithmetic supplies the transcendental input bounds, including the Dickman and exponential quantities. The final signed reductions retain outward error bounds as well [6].

At this point an arithmetic defect invalidated an earlier set of cap enclosures. I had relied on negative polynomial division to implement an upper ceiling. The division convention did not have the required coefficientwise behavior, and tiny positive upper coefficients could disappear. Increasing precision would not repair a wrong direction of rounding. The correction was to use the elementary nonnegative integer identity

$$\left\lceil \frac{n}{Q} \right\rceil = \left\lfloor \frac{n+Q-1}{Q} \right\rfloor, \qquad n \ge 0,$$

on every coefficient, including internal zero coefficients. The earlier enclosures had to be discarded and the comparison repeated. Sparse products with tiny coefficients were especially important checks, since large smooth examples could hide this failure.

This is a rounding error in the wrong direction, so the earlier narrow intervals do not enclose the quantities I need. After the correction, some low-precision bounds become much wider: the tiny upper coefficients are now being retained. I must recover sharpness from the calculation itself, while preserving the outward bounds.

There was another large improvement that changed only the organization of the exact bound. Distributing a Chernoff exponential across all coordinates created huge and tiny intermediate quantities. On a fixed total-index row, their product is known exactly. I could leave the local factor $e^{-\theta L_{<l}}$ in the fragment laws, perform the positive convolution, and apply the common radial exponential afterward. The full-cell upper displacement still had to be included. This factorization avoids unnecessary numerical range without weakening or changing the majorant.

Likewise, a polynomial with a known initial string of zero coefficients can be represented as $z^v \widetilde{P}(z)$. The shift v is exact, so multiplying the shorter positive polynomials and restoring the shifts gives the same enclosure. These simplifications were worthwhile because the proof required all the source components, including very small ones. A negligible-looking component could not simply be omitted from the cover.

The first complete comparisons also corrected my expectations about grid size. At 65,536 intervals, the corrected cap-only surplus was positive, but the accumulating source upper bounds used too much of it. The coarse-grid source experiments had underestimated how the normalized dominant components would behave on the finer fixed trial. I did not interpret this as a failure of the candidate theorem; it was a failure of that particular certified comparison to leave enough room.

The final calculation used 98,304 intervals. The largest-prime bins in the dominant outer $H_{5/2}$ role received extra resolution where their contributions were largest. Conversely, some small adjacent low-witness intervals could be replaced by their exact union and enclosed again. Such a merge is a new upper-bound calculation. It is not justified by adding together earlier small numerical observations and assuming that a broader Chernoff envelope will remain small. Indeed, merging all the low intervals of an apparently minor inner role gave a very poor bound. The exponential penalty can grow much faster than the saved number of intervals suggests.

The accepted outer H_2 merge was modest and explicit:

$$(l, u] = \left(\frac{1}{20}, \frac{324}{3125}\right], \qquad \theta = 120.$$

It replaced four consecutive intervals by their exact union, while retaining the first two and the later dominant intervals. The outer $H_{5/2}$ largest-prime partition used twelve bins, obtained by bisecting the first four of eight original intervals and retaining the remaining four. The final component counts were

	low witnesses	two large fragments	higher ranks
O_2	10	6	1
$O_{5/2}$	22	12	1
$I_{0,2}$	5	1	1
$I_{0,5/2}$	7	2	1
$I_{1,2}$	6	4	1
$I_{1,5/2}$	11	5	1

Their sum is 97. Each row refers to a particular physical role and obstruction order, rather than to a generic probability of a bad fragment. All intervals have rational endpoints, and each higher-rank term covers the entire remaining factorial tail.

The final arithmetic comparison used

$$\rho_* = \frac{2624989}{10^7}.$$

The cap-only quotient was bounded below by

$$\frac{\rho_* J_{\lambda,\mathrm{cap}}}{I_{\mathrm{cap}}} \geq 1.000206079418044888,$$

and the complete source correction, divided by the denominator lower bound, was at most

$$\frac{\mathcal{E}_O + \beta}{I_{\mathrm{cap}}} \leq 0.000696075048551029.$$

The correction is multiplied by ρ_* in the final comparison. Even these displayed, safely rounded numbers leave a positive surplus. Recombining the underlying exact rational endpoints gives the sharper normalized margin

$$\boxed{\frac{\rho_*(J_{\lambda,\mathrm{cap}} - \mathcal{E}_O - \beta) - I_{\mathrm{cap}}}{I_{\mathrm{cap}}} \geq 0.000023360484159866 > 0.}$$

The last decimal is the result of that exact endpoint comparison; it is not obtained by treating the preceding printed decimals as exact values.

The arithmetic target can now be stated without any further optimization. The tuple is

$$\begin{aligned}\mathcal{H} = \{&0, 2, 6, 12, 20, 26, 30, 32, 36, 42, 48, 50, 56, 60, 68, 72, 78, 86, 90, 92,\\ &98, 102, 110, 116, 120, 126, 132, 138, 140, 146, 152, 156, 158, 162, 168, 170,\\ &176, 180, 182, 186\}.\end{aligned}$$

It has forty elements and diameter 186. Admissibility is a finite check for primes at most 40; for larger primes, forty residues cannot fill every class. For example, missing residue classes for the small primes are

p	2	3	5	7	11	13	17	19	23	29	31	37
missing class	1	1	4	3	7	5	11	8	16	9	4	3.

The finite variational inequality is now in place. Together with the distribution estimate for these moduli, the signed minorant comparison, and the hereditary-support and finite-band passages, it gives positive excess over the contribution of a single prime. Thus infinitely many translates of this admissible tuple contain at least two genuine primes, by the multidimensional Selberg criterion [1, 2].

Any two such primes have distance at most 186, and the interval between them contains a pair of consecutive primes with no larger gap. This is the point at which the source argument concludes

$$\boxed{\liminf_{n\to\infty}(p_{n+1}-p_n) \leq 186.}$$

References

[1] J. Maynard, *Small gaps between primes*, Ann. of Math. (2) **181** (2015), 383–413.

[2] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. **1** (2014), Art. 12; bibliographical erratum, Res. Math. Sci. **2** (2015), Art. 15. Numbered references use arXiv:1407.4897v4.

[3] R. C. Baker and A. J. Irving, *Bounded intervals containing many primes*, Math. Z. **286** (2017), 821–841. Numbered references use arXiv:1505.01815v1.

[4] J. Stadlmann, *On primes in arithmetic progressions and bounded gaps between many primes*, Adv. Math. **468** (2025), Art. 110190. Numbered references use arXiv:2309.00425v3.

[5] D. H. J. Polymath, *New equidistribution estimates of Zhang type*, Algebra Number Theory **8** (2014), 2067–2199. Numbered references use arXiv:1402.0811v3.

[6] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.

[7] G. Sarajian and A. Weingartner, *An extension of smooth numbers: multiple dense divisibility*, J. Number Theory **280** (2026), 278–317.